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13 Ligation of Bleeding Ulcer, Antrectomy, Vagotomy, and Gastrojejunostomy

Bruce Schirmer

INDICATIONS/CONTRAINDICATIONS

Bleeding duodenal ulcer is a surgical emergency that has decreased in frequency over the past several decades due to the overall decrease in the incidence of peptic ulcer disease. However, the condition still arises and is a life-threatening one for the individual who develops it. Surgery remains an important treatment option for the patient with a bleeding duodenal ulcer. Its role has been decreased from primary line therapy in most situations to secondary treatment when the patient is stable enough to undergo less invasive therapy, in particular flexible endoscopic attempts to relieve the bleeding.

Indications

Specific indications for the performance of antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer are as follows:

1. Initial treatment of a patient with a bleeding duodenal ulcer causing hemodynamic changes unresponsive to intravenous fluid and blood product transfusion.
2. Treatment of a patient with bleeding duodenal ulcer who has failed attempts at endoscopic therapy to control the bleeding ulcer and for whom interventional angiographic therapy to control the bleeding is either unavailable, has failed, or is felt to be contraindicated.
3. Treatment of a patient with bleeding duodenal ulcer for whom no endoscopic or radiographic therapeutic options to arrest the bleeding are available.

In addition to the above three indications for the operation, the following parameters are also considered important by many surgeons to be present to warrant this operation:

1. The patient should have had a history of previous peptic disease to warrant resectional therapy.
2. The previous peptic disease should not have been so severe as to render the pyloric and postpyloric area severely scarred.

3. The patient's condition in the operating room must be one of hemodynamic stability once the bleeding ulcer has been oversewn.

Contraindications

Specific contraindications for the performance of antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer are as below. This list may not be totally comprehensive but should encompass the major contraindications.

1. Hemodynamic instability in the operating room necessitating the performance of the most rapid operation feasible (vagotomy, pyloroplasty, and oversewing of the ulcer).
2. Severe scarring of the pyloric and proximal duodenum, such that the resection staple line for the distal margin of the antrectomy would be at jeopardy for breakdown.
3. No previous history of peptic disease or potential other etiologies (e.g., aspirin or nonsteroidal anti-inflammatory drug ingestion) that may have contributed to the ulcer diathesis.
4. Previous antireflux surgery, placement of lap-adjustable band for weight loss, or other operation with extensive dissection around the gastroesophageal junction which would contraindicate easy performance of a truncal vagotomy.

PREOPERATIVE PLANNING

The overwhelming issue regarding preoperative planning for the operation is the attention to achieving hemodynamic stability in the patient and the means to monitor such stability and to provide vigorous transfusion volume should it be necessary. For any major gastrointestinal bleed, the following resuscitative and patient care measures should be instituted as the first priority of care:

1. Transfer the patient to an ICU setting where continuous monitoring of vital signs is performed
2. Central line placement for both rapid fluid resuscitation and for monitoring volume status
3. Foley catheter to measure organ perfusion and hence resuscitation success
4. Availability of appropriate blood products should the bleeding worsen or continue unabated. These could include, as most appropriate, whole blood, fresh frozen plasma, packed red blood cells, and if significant transfusion requirements occur, platelets and cryoprecipitate
5. Reversal of any potential anticoagulant medication
6. Large bore intravenous access for transfusion requirements

Confirming the diagnosis of bleeding duodenal ulcer is normally done using flexible upper endoscopy. Usually endoscopic measures to control the bleeding are performed at the time of the procedure. Lack of efficacy of such measures is considered an appropriate indication to proceed with surgical intervention. Performing an extensive operation such as antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer is normally not performed without a clear diagnosis of bleeding duodenal ulcer. It is feasible that such a diagnosis can be inferred by the clinical picture but not confirmed endoscopically. In such situations, endoscopic success could have been limited by excessive blood in the lumen of the duodenum or due to other technical reasons. On occasion, diagnosis by angiography with inability to perform angiographic measures to stop the bleeding or failure of such measures could also serve as an appropriate confirmation of the diagnosis of bleeding duodenal ulcer and the need to perform surgical therapy.

Prior to surgery, a thorough history of the patient's previous problems with gastrointestinal issues, including any history of previous ulcer disease or long-standing symptoms consistent with ulcer disease, is appropriate. Given such a history, the use of an antrectomy with vagotomy, rather than just vagotomy and drainage procedure, for the treatment of complications of duodenal ulcer is more justified.

Surgical Procedure

The operation is, due to its nature as an emergency procedure for life-threatening bleeding, normally conducted as an open operation through an upper midline incision. Its performance as a laparoscopic operation could be feasible, but the two major factors that would make laparoscopy more difficult are the maintenance of hemodynamic stability in the setting of creating a pneumoperitoneum and the difficulty in suctioning the actively bleeding area in order to place the sutures properly to oversew the ulcer without losing the pneumoperitoneum and the operative field of vision.

Oversewing the Ulcer

Once an upper midline incision is made and the abdominal organs are made accessible, the first priority and hence also first step of the operation is to perform the oversewing of the ulcer. The steps of this portion of the operation are as follows:

- The duodenum is exposed. A Kocher maneuver may assist in bringing the duodenum up into the field and should be used.
- The pylorus is identified.
- An anterior duodenal incision is made starting 1 cm beyond the pylorus, and extending 2 cm into the duodenum (Fig. 13.1).
- The bleeding site should be within visualization using this access. On occasion, further extension of the duodenal end of the incision may be needed.
- The bleeding ulcer is normally on the posterior surface of the duodenum, in the location of the gastroduodenal artery. The course of the artery is usually through the distal first part of the duodenum or most proximal second portion.
- The ulcer bleeding vessel is now directly oversewn with three simple silk sutures. Other materials may be used, but the suture weight must be heavy enough to allow it to be tied to gather the ligated tissue together without breaking. A 2-0 weight suture is recommended.

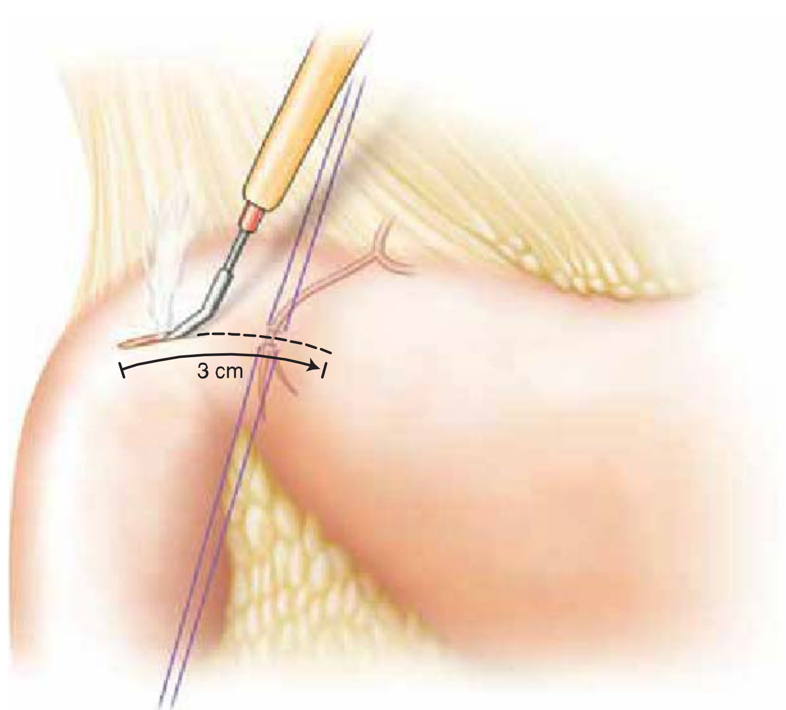

Figure 13.1 Performing an incision 1 cm distal to the pylorus and extending it 2 cm distally in the duodenum to expose the site of the bleeding ulcer.

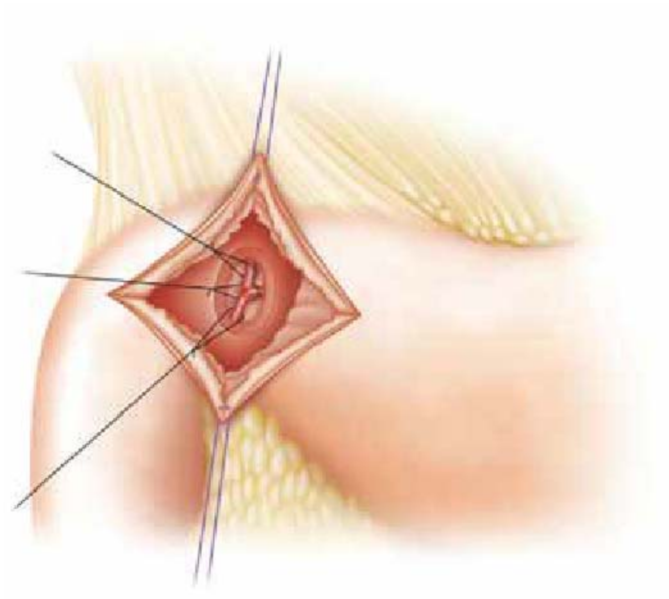

Figure 13.2 Placement of sutures for oversewing of a bleeding duodenal ulcer. Three sutures are required. One is placed proximally and one distally through the axis of the bleeding vessel. The third is placed at a 90-degree orientation medially to the gastroduodenal artery in order to ligate the frequently present medial pancreatic branch of the artery.

- The sutures are placed as illustrated in Figure 13.2. One is at right angle to the course of the artery proximal to it, a second in that same orientation distal to it, and the third at a 90-degree orientation on the medial or proximal side of the bleeding to occlude the frequently present transverse pancreatic branch of the gastroduodenal artery.
- Once the bleeding is controlled, the duodenotomy is closed with a running layer of absorbable suture.

Performing the Antrectomy

The antrectomy is now performed once it is confirmed that the most proximal duodenum is not excessively scarred and therefore is amenable to surgical closure. The steps of the antrectomy are as follows:

- The plane underneath the surface of the most proximal duodenum is dissected to allow passage of a stapler through it. Small vessels in this area may need division with an ultrasonic scalpel.
- The GIA-type stapler is fired across the most proximal duodenum at this location. The staple line should be beyond the pylorus. It should also be proximal to the closure of the duodenotomy whenever feasible (Fig. 13.3). Inclusion of the suture closure

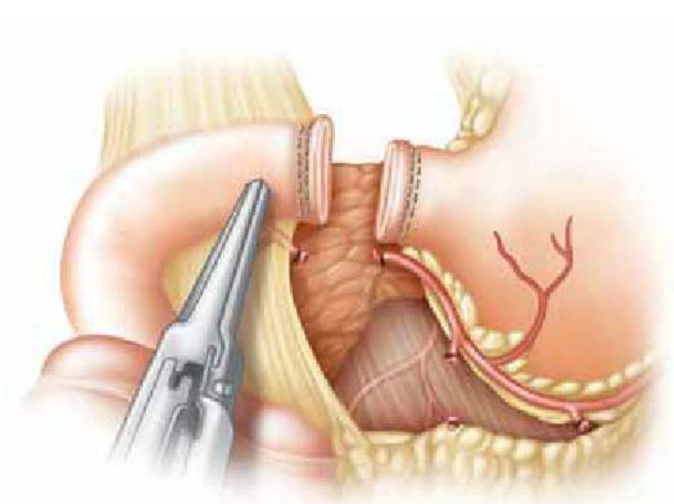

Figure 13.3 A GIA-type stapler is fired across the proximal duodenum just beyond the pylorus.

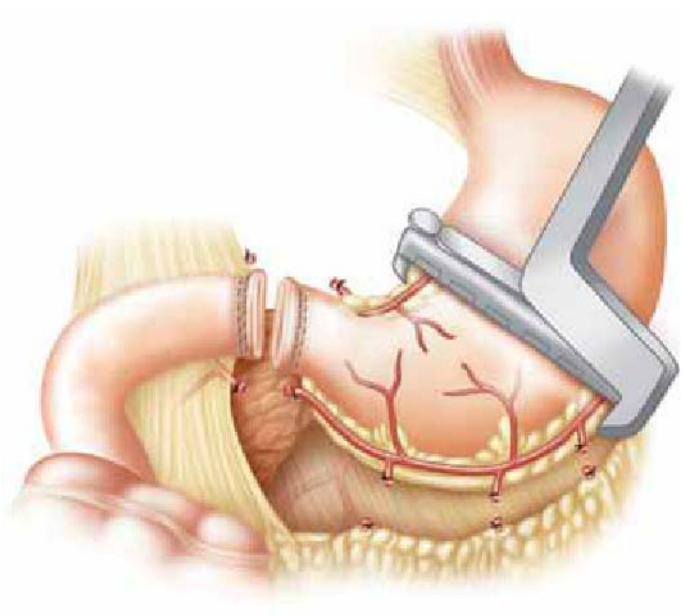

Figure 13.4 Dividing the stomach with a TA-type stapler to perform the antrectomy.

of the duodenotomy will weaken that suture line, and if this occurs, the suture line must be reinforced to prevent disruption.

- The site of proximal division of the stomach is determined. This is normally at the incisura on the lesser curvature of the stomach and in a point directly radially opposite to it on the greater curvature of the stomach. These sites are marked by local division of vessels adjacent to the lesser and greater curvatures of the stomach with the ultrasonic scalpel.
- The vessels of the right gastroepiploic artery running adjacent to the greater curvature of the stomach along the antrum are divided using either suture ligation technique or with the ultrasonic scalpel. Special care is taken to ligate the main trunk of the right gastroepiploic artery.
- The vessels along the lesser curvature of the stomach, beginning at the pyloric division, are also ligated in a similar fashion. Here the right gastric artery must be securely identified and ligated.
- The stapler is used to divide the stomach at the previously determined sites of division (Fig. 13.4). Care must be taken to be sure no nasogastric or other tubes are in the lumen of the stomach. The blue, gold, or green load of the TA-type stapler is used, depending on the thickness of the gastric tissue.
- Both the gastric staple line and the duodenal staple line are inspected for integrity and hemostasis. Simple sutures are used if any areas of concern are noted.

Performing Vagotomy

A truncal vagotomy is indicated in this setting. Optimal elimination of the cholinergic stimulation of gastric secretion is sought; hence complete or truncal vagotomy is performed. The steps of the vagotomy are as follows:

- The phrenoesophageal ligament is divided anteriorly to expose the anterior surface of the esophagus. Dissection of the peritoneum is extended to the left to the angle of His and to the right to just past the right crus of the diaphragm.
- The anterior vagus nerve is located buried within the musculature of the anterior distal esophagus (Fig. 13.5).
- A 1 cm or longer segment of the anterior nerve is resected, with small hemoclips placed on the proximal and distal resected ends of the nerve. These clips serve to achieve hemostasis but also serve as a radiographic marker to demonstrate the performance of a truncal vagotomy.

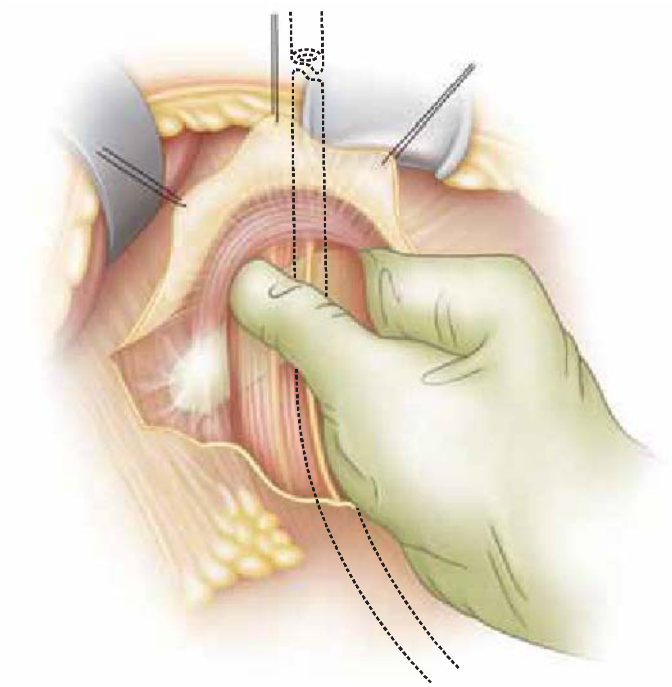

Figure 13.5 Locating the anterior vagus nerve which is buried in the anterior esophageal muscle wall.

- The posterior vagus is clearly identified. It often lies posterior to the esophagus, along the surface of the crura (Fig. 13.6). **Failure to clearly identify the posterior vagus nerve is a common technical error of this procedure.** If the nerve is not clearly palpable along the posterior surface of the esophagus, the surgeon must extend the plane of blunt dissection with a finger up along the patient's right side of the esophagus until the nerve is clearly palpated. Then it can be traced distally to assure proper identification of the posterior nerve at the hiatus.
- The posterior nerve is similarly resected and clipped as was the anterior nerve.

Performing Gastrojejunostomy

Gastrojejunostomy is performed as the preferred drainage method of the stomach after antrectomy. The creation of a single anastomosis is preferred. Resection of just the antrum

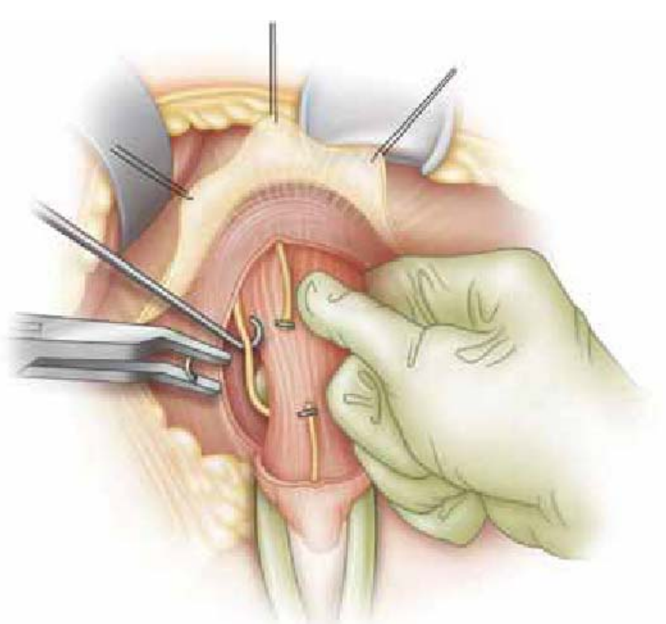

Figure 13.6 Locating the posterior vagus nerve, which is often not closely attached to the posterior esophageal wall, but instead courses more posteriorly, lying atop the area of intersection of the diaphragmatic crura.

will normally allow adequate gastric volume and peristalsis to prevent reflux of jejunal secretions into the esophagus. The key steps in this portion of the operation are as follows:

- Identification of the ligament of Treitz to identify the most proximal jejunum. The jejunum just distal to this is then brought through an opening in the transverse colon mesentery just to the patient's left of the middle colic vessels. The gastrocolic ligament is opened widely to allow good access to the posterior surface of the stomach.
- The jejunal loop is secured to the posterior surface of the distal stomach, just above the resection line, with several tacking sutures of 3-0 silk. Isoperistaltic alignment of the jejunum is ideal. In aligning the jejunum, however, the most important aspect is to avoid a kink and hence obstruction of the course of the bowel as it traverses up to and back from the anastomosis with the stomach.
- Use of a double-staple technique decreases the chance of kinking of the distal or outflow portion of the anastomosis. The midportion of the segment of jejunum that is brought up to the surface of the stomach is used for the creation of the anastomosis. This point must be at least 4 cm proximal to the distal end of the resected stomach to allow for double stapling. A gastrostomy and a corresponding enterotomy at that location are created (Fig. 13.7).
- The linear stapler, usually a 45-mm blue load, is fired first in one direction (Fig. 13.8) and then in the opposite direction through the same opening (Fig. 13.9).
- The defect from the stapling is now closed with a single layer of absorbable suture.
- Tacking sutures are placed 1 cm beyond the end of the anastomosis both proximally and distally (Fig. 13.10). These sutures are important to prevent outflow or inflow obstruction to the gastrojejunostomy. They are more important than making a very long gastrojejunostomy, since no matter what the length of the anastomosis, its emptying is eventually determined by the caliber of the lumen of the efferent limb of jejunum. If this is kinked, as may often occur without the tacking suture, then outlet obstruction occurs. **Failure to prevent outflow obstruction of the gastrojejunostomy is the most common technical error of this operation.** If the outflow obstruction is severe enough, the pressure from the obstruction will distend the duodenal limb and result in blowout of the stapled end of the duodenal stump.

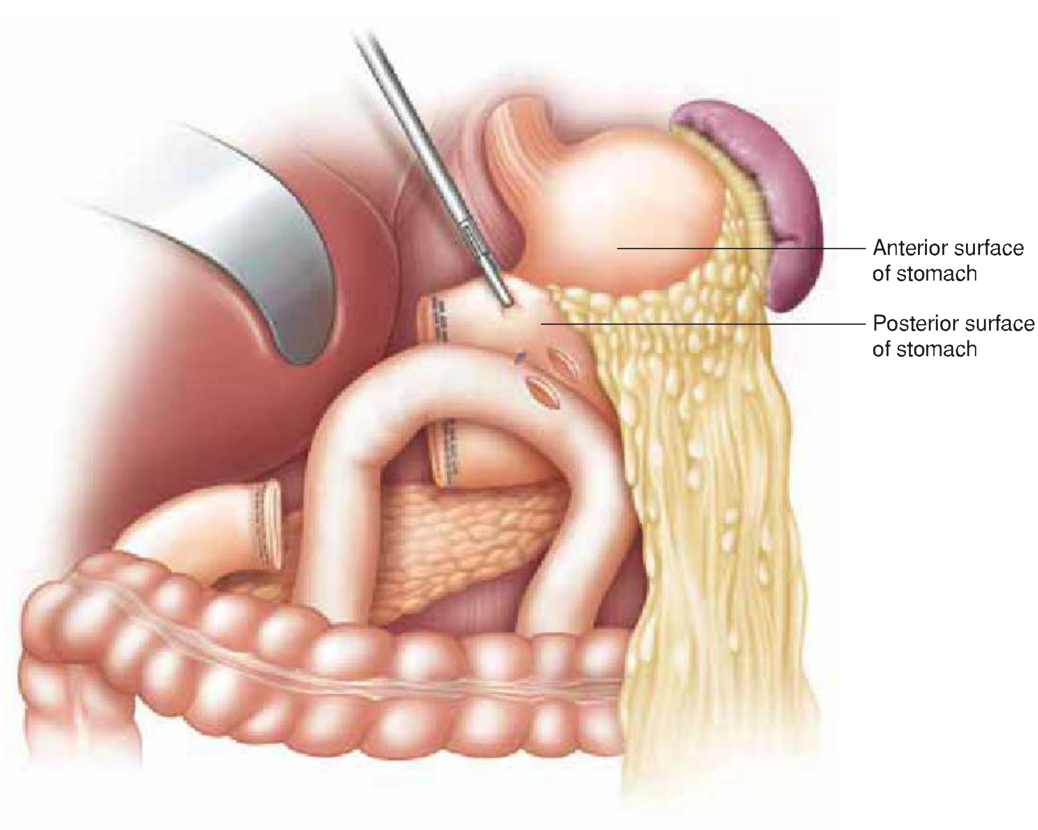

Figure 13.7 Making a gastrostomy and an enterotomy to begin stapling the gastrojejunostomy. Note the jejunum is aligned along the posterior surface of the distal stomach to facilitate emptying.

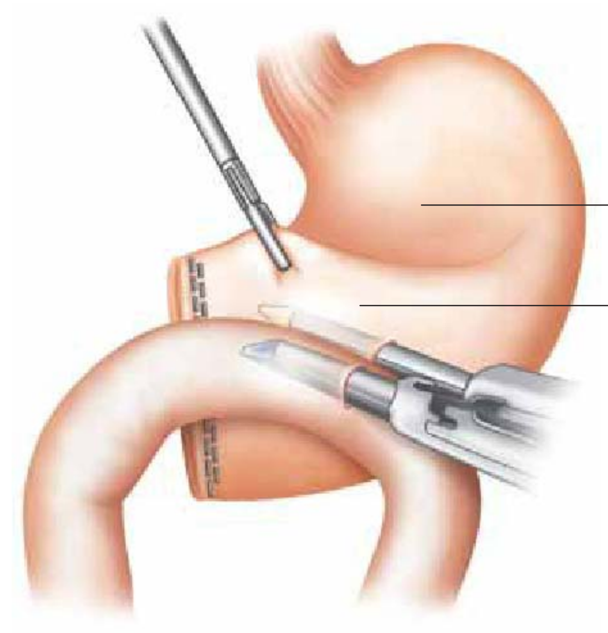

Figure 13.8 Firing the GIA-type stapler in one direction to begin the gastrojejunostomy.

- The afferent and efferent limbs of the jejunum are tacked to the transverse colon mesentery to prevent another loop of intestine from migrating upward into the lesser sac. They may also be tacked to each other at this location for additional security.

POSTOPERATIVE MANAGEMENT

The postoperative care of the patient with a bleeding duodenal ulcer, who has undergone antrectomy and vagotomy as a curative operation, must be focused on the continued resuscitation of the intravascular space, with adequate intravenous fluid administration.

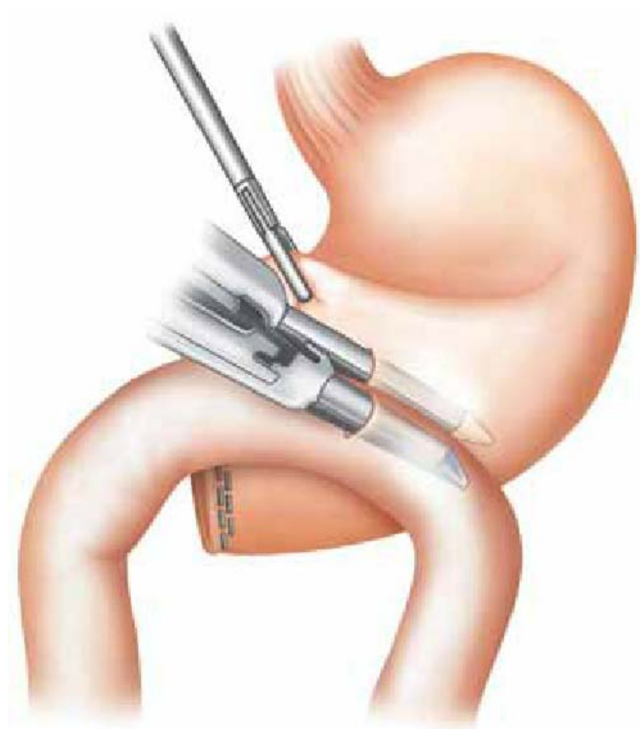

Figure 13.9 Firing the GIA-type stapler in the other direction to perform a double-stapled gastrojejunostomy anastomosis. This technique will decrease the incidence of postoperative outflow obstruction of the gastrojejunostomy.

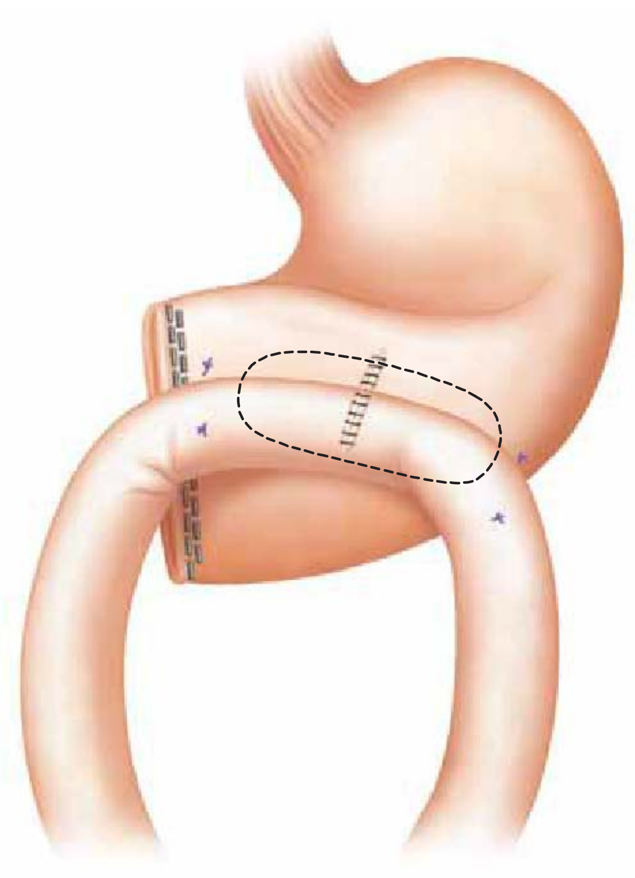

Figure 13.10 Placing tacking sutures 1 cm from the ends of the anastomosis both proximally and distally such that the jejunum is not kinked either in its inflow or outflow from the area of the gastrojejunostomy.

Hemodynamic stability, good perfusion of organs, and no signs of organ dysfunction from the stress and potential shock of the bleeding episode are the goals of therapy at this point. Careful monitoring of organ function, preferably in an ICU setting for 24 to 48 hours, is indicated. If a significant period of hypotension occurred during the bleeding, then the surgeon must be alert for signs of renal, pulmonary, neurologic, cardiac, or other organ injury secondary to the period of hypotension. Replenishment of the hemoglobin-carrying capacity of the blood to an adequate level is also indicated. This level can vary based on age and underlying medical conditions. Signs of organ dysfunction require continued treatment in the ICU setting with therapy focused on restoration of full organ function. These goals and the major other postoperative care issues and treatment are as follows:

- Monitoring for no further signs of bleeding or other hemodynamic instability
- Restoration of full vital organ function
- Restoration of adequate hemoglobin capacity
- Monitoring for adequate gastric emptying (signs of nausea, vomiting, distention are the signal that this may be an issue)
- A nasogastric tube is not indicated unless the patient does develop postoperative gastric ileus or partial outlet obstruction. Placement is based on clinical and potentially also radiographic criteria.
- If poor gastric emptying occurs, decompression with a nasogastric tube is indicated if it is severe. Otherwise, transient use of intravenous fluids and nutrition until gastric emptying is improved may be all that are necessary.
- A postoperative gastrografin swallow is variably used by surgeons to assess for gastric emptying as well as for security of all staple lines and anastomoses prior to initiating oral intake.
- Mild gastric dysfunction after vagotomy is usually the rule. The patient should be counseled that poor food tolerance may be present initially, but this will improve. On occasion pharmacologic therapy such as erythromycin may be used.

- Treatment of the underlying ulcer diathesis has been largely accomplished by the operation, but the patient should be tested for *Helicobacter pylori* presence in the resected stomach specimen. In many cases, surgeons will empirically give the patient a course of triple therapy for treatment of *H. pylori*. Subsequent hydrogen breath testing to confirm eradication of the organism is indicated.
- The patient must be counseled about postoperative dumping and diarrhea, not uncommon after vagotomy and antrectomy. These usually occur in tandem, though may be manifested individually. Dietary measures will alleviate most symptoms of dumping and diarrhea. Avoidance of highly concentrated carbohydrates and fats is the hallmark of such diet modification. Slower eating and lower volumes of food at a sitting improve these problems as well.
- Past experience has shown that, with the proper performance of antrectomy, truncal vagotomy, and gastrojejunostomy, the recurrence of ulcer is in the 1% range. Most surgeons do not recommend prolonged proton pump inhibitor therapy for these patients. However, recurrence of peptic symptoms may warrant their reinstitution.

COMPLICATIONS

Some of the most severe complications which may be seen after this operation have already been alluded to in the above text. These complications may be thought of as constituting four general areas. They are as follows:

- Overall typical complications that one may see following any abdominal laparotomy.
- Sequelae and complications from the period of gastrointestinal hemorrhage, particularly episodes of hypotension and organ ischemia that may have occurred. This group of complications is the most life-threatening.
- Complications specific to the operation. These would further be broken down into short-term or long-term complications.
- Potential complications that may arise if the ulcer disease were to recur or persist.

Overall complications that one would anticipate after a laparotomy include atelectasis, wound infection, venous thromboembolism, and postoperative ileus. Cardiovascular events would be less anticipated but not rare. Attention to postoperative pulmonary toilet, early ambulation, and adequate pain control without excessive sedation all are paramount to combating the complication of postoperative atelectasis. Appropriate prophylactic antibiotics administered parenterally before the performance of the incision are known to have the greatest impact on decreasing postoperative wound infection. Broad-spectrum antibiotics effective against both gram-positive and some gram-negative bacteria are used (cefazolin is the usual choice). However, careful tissue handling, hemostasis, irrigation, and minimization of wound contamination are all also important in minimizing wound infection rates.

Complications which occur after an episode of hemorrhagic shock may include organ dysfunction from any and all organs sensitive to the effects of the period of hypotension. The most frequently affected are the renal and pulmonary systems, with acute tubular necrosis/acute renal failure and acute respiratory distress syndrome resulting from this injury. Cardiac ischemia may result, leading to some degree of myocardial infarction if severe enough. Neurologic sequelae may arise, especially if pre-existing cerebrovascular disease is present. Cholestasis and acute hepatic injury is common but is usually more reversible than others due to liver capacity. Pre-existing liver disease, however, could result in acute decompensation of liver function. Treatment for all these conditions rests in now what is generally much better understood by intensive care specialists: measures to restore perfusion, maintain organ viability, minimize infectious complications, and support organ function until recovery occurs.

Complications specific to the operation include short-term and long-term complications. For these complications, it is also relevant to discuss the most common measures known to successfully prevent them. The short-term commonly expected complications include the following:

- Persistent or recurrent bleeding from the ulcer: This would be a technical failure of the oversewing and would potentially require reoperation to place additional sutures to achieve hemostasis. Careful attention to suture placement and correct technique are the best measures to avoid this complication.
- Bleeding from a staple line or anastomosis: This is best avoided by observing all staple lines at the time of surgery for hemostasis. If wider staple loads (such as green cartridges) are used to divide the stomach, consideration for oversewing the staple line should be given, since such staple sizes are more prone to postoperative bleeding.
- Gastric outlet obstruction: This is a common technical error caused by improper attention to the outflow tract of the gastrojejunostomy. It must be secured in a position such that the bowel will not kink and collapse the lumen of the opening. Placement of the anastomosis on the posterior surface of the stomach facilitates emptying. There should be a smooth curvature of the intestine both going up to and exiting away from the posterior surface of the stomach and the gastrojejunostomy.
- Breakdown of the duodenal stump: This is perhaps the most feared complication of this operation and one which is associated with significant morbidity and mortality. There are two major causes. The first is that the closure is technically inadequate. The second is that distal obstruction creates enough pressure for the staple line to rupture. Treatment is adequate drainage and intravenous antibiotics, which may or may not involve reoperation.

Complications from recurrence of the ulcer disease include recurrent abdominal pain, possible recurrent bleeding, and gastric outlet obstruction from marginal ulceration. Confirmation that a complete vagotomy was performed is one step in treatment; reoperation is needed if it was incomplete. Medical therapy is otherwise usually adequate to overcome recurrence of ulcer diathesis when it occurs. Rarely a neoplastic (gastrinoma) cause may be responsible.

RESULTS

This operation is done primarily for patients with urgent or emergent bleeding problems. They often have associated hypotension and are critically ill in most situations. Therefore, the outcomes of this procedure must be judged by the clinical setting under which it is performed and the patient population for which it is performed. Most results published in the literature date back to the days when surgery for ulcer disease was more common, the prevalence of ulcers was more common, and medical treatment of ulcer disease was not as effective. The operation described in this chapter is now very selectively used by surgeons for the treatment of patients with bleeding duodenal ulcer.

The literature, and practical experience, has shown that patients who are poor candidates to begin with for surgery have multiple medical problems or who have become *in extremis* from a prolonged period of shock and hypotension from gastrointestinal bleeding are likely to have a very high mortality rate from surgical therapy. Furthermore, for patients who are at high risk for surgery, a resectional operation such as antrectomy would be relatively contraindicated in favor of a less involved operation such as simple oversewing of the ulcer with pyloroplasty and possibly truncal vagotomy.

Most reports of surgical therapy in the literature combine a variety of operations and a variety of degree of severity of illness in patients in their reports.

Historically speaking, the generally quoted mortality rate overall for patients with hemorrhage from duodenal ulcer disease is in the 12% range. First-line therapy for such bleeding patients is normally endoscopic therapy, which has an expected success rate of 85% to 95% of stopping the bleeding. This figure is influenced by local expertise.

The success of endoscopic therapy for rebleeding, or failed first-time therapy, is lower at about 58% to 72%.

The literature has shown that overall surgical therapy and repeat endoscopy after an initial failed endoscopic treatment have roughly equivalent success rates and mortality rates. The decision to perform surgery at the time of a failed initial endoscopic therapy is the judgment of the surgeon and again is based on local expertise, patient condition, and other qualifying factors. Antrectomy and vagotomy with gastrojejunostomy could be used in this setting, but it would generally be applied to more fit patients with less transfusion requirements and greater hemodynamic stability. Factors that predict failure of endoscopic therapy include

- initial hypotension
- an ulcer size greater than 2 cm in diameter.
- vessel size of 1 mm in diameter or larger

Initial endoscopic findings of a large duodenal ulcer with a visible bleeding vessel of 1 mm or greater would likely lead to surgery if any sign of rebleeding did occur or if the initial endoscopic therapy was unsuccessful.

If endoscopic therapy fails, then the surgeon faces the option of performing salvage surgery or, more recently, referring the patient for a transcatheter arterial embolization (TAE) therapy. A recent comparison of results of TAE versus surgical therapy was published by Venclausakas et al.. This report showed that the mortality for such patients was comparable at 21% for the TAE group of patients and 22% for the surgical patients. The rebleeding rate was higher, but not statistically so, for the TAE group, which is not unexpected. Though overall mortality for the group was comparable, the authors noted the TAE group had a higher age and Apache II score, meaning they were overall poorer risk candidates. For patients with an Apache II score of 16.5 or greater, the TAE mortality was 23% versus the surgical mortality of 50%.

Another review by Eriksson et al. showed that in such a situation the mortality for TAE was less than for surgery, at 3% vs. 14%. In that surgical group, 35 of 51 operations involved resection, whereas only 14% involved simple oversewing of the ulcer. Use of the latter approach may have decreased the surgical mortality. Also, the average age of the patients who died after surgery was 80, suggesting operative therapy in this elderly a patient population may not have been the best choice versus TAE.

Certainly the surgeon should be the ultimate decision maker in the choice of therapy for bleeding duodenal ulcer. Lower risk patients are probably better treated earlier with surgical therapy. A history of peptic ulcer disease, previous bleeding, and hemodynamic stability are the optimal settings under which to perform antrectomy, vagotomy, oversewing of the ulcer, and gastrojejunostomy. However, reports of a pure series of such patients alone are difficult to find in the literature, and those that exist are very dated and do not reflect improvements in modern surgical perioperative care, critical care, patient resuscitation, and improved antibiotic and medical therapy. Given the traditional overall mortality of surgical therapy for bleeding ulcer as being approximately 12%, selected patients today should be expected to have improved outcomes over that figure.

The difficult patient for the surgeon becomes the patient who has failed endoscopic therapy and then the decision of whether to perform salvage surgical therapy or TAE must be determined. The role of TAE is largely influenced by the local expertise of that group of radiologists in a given institution. However, the overall literature results show that

- TAE is technically successful in over 90% of cases
- control of hemorrhage is between 50% to 90% in case series in TAE
- rebleeding after TAE is reported to occur between 8% and 40% of cases
- higher mortality is associated with rebleeding in TAE
- overall mortality for patients with bleeding duodenal ulcer treated by TAE has been reported as being between 0% and 25%
- mortality is less if TAE is successfully performed early in the course of the bleeding, as would be expected

A recent review of the experience at a large teaching hospital reviewing the incidence of peptic ulcer disease showed that in the decade from 1995 to 2004, the incidence of surgery for gastric or duodenal ulcers decreased from 6.7% to 3.8% of cases, with the gastric ulcer group representing the major decrease. The incidence of surgery for duodenal ulcer was constant at about 7.5% of cases. During the decade, the incidence of using an acid-reduction operation such as is described in this chapter for the operative treatment of bleeding duodenal ulcer fell from 50.6% to 31.6% over that time frame, showing that recent trends are to just perform an operation focused on stopping the bleeding and not reducing acid secretory capacity. The overall results of surgical therapy for patients with gastric or duodenal ulcers was a mortality of 6.0%, morbidity of 23.9%, and an average length of hospital stay of 15.9 days.

The shift away from performing an acid-reducing operation was in part borne out by studies that examined the effectiveness of using medical therapy to eradicate *H. pylori* as part of the therapy to treat duodenal ulcer disease and to prevent recurrent duodenal ulcer disease. Duodenal ulcer disease is felt to occur as a result of a hypersecretory state of acid production, stimulated in large part by antral D-cell dysfunction secondary to *H. pylori*-induced antral gastritis, which in turn limits somatostatin release and exacerbates acid production. Such acid production in the face of *H. pylori* colonization of the duodenal mucosa leads to ulceration. A study by Ng and associates showed that appropriate treatment of *H. pylori* with four drug medical therapy at the time of closure of perforated duodenal ulcer resulted in a 5% ulcer recurrence rate over the next year versus 38% recurrence for patients who did not receive such therapy against *H. pylori*. The ability to thus treat potential ulcer recurrences in the future for patients with previously unknown peptic ulcer disease have led in large part to the shift in surgical thinking regarding treatment for patients with duodenal ulcer, with a clear trend away from acid-reducing operations to simple closure of perforation or oversewing of bleeding ulcer. Patients who are candidates for acid-reducing operations would be those for whom risk factors still exist or cannot be diminished:

- chronic refractory nonsteroidal anti-inflammatory drug ingestion
- smoking
- noncompliance with medical therapy for *H. pylori* or acid secretion

Salvage surgery is often one of the main indications for performing surgery for bleeding duodenal or gastric ulcer in recent years. In this situation, where patients are often quite critically ill, mortality from surgery or any therapy can be very high. Walsh et al. described a series of 50 patients who had failed endoscopic therapy (an average of over two procedures each), many of whom developed their GI bleeding after hospital admission while in an intensive care setting. The Apache scores of 79 and the 64% incidence of organ failure as well as the average transfusion amount of 24 units of blood puts these patients in a very high risk for death from any treatment. In this group, 12 patients underwent salvage surgery for bleeding duodenal ulcer, with a resection performed in seven and a simple oversewing of the ulcer in five. The mortality for such surgery was 50%.

CONCLUSIONS

The performance of antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer is now a relatively rarely performed operation due to improved medical therapy against the causative agent of most duodenal ulcers, *Helicobacter pylori*. Nevertheless, when it is indicated, surgeons must be familiar with the procedure and adequately skilled to perform it to produce good outcomes. The patients who are potential candidates for such an operation may be threatened by significant duodenal hemorrhage. Therapeutic endoscopy is the first-line treatment for bleeding duodenal ulcer in most settings. Failure of such therapy may then be an indication for the performance of surgical therapy. Performing an antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer is indicated largely for those patients who have other risk factors for duodenal ulcer disease other than simple presence of *H. pylori,* or who have a previous history of such ulcers, and who are hemodynamically stable enough and good surgical candidates to undergo a major resective operation. For most patients with bleeding duodenal ulcer and no other risk factors, simple oversewing of the ulcer along with adequate medical therapy for *H. pylori* has proven to be appropriately effective for most patients. When any surgery is used for salvage after failure of multiple endoscopic therapies and/or transarterial catheter embolization therapy, the mortality of such operations is high. Use of antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer in such settings probably should be rarely done in favor of a more simple operation to oversew the bleeding ulcer.

When antrectomy, vagotomy, gastrojejunostomy, and oversewing of a bleeding duodenal ulcer is performed, careful attention to technical considerations outlined in the chapter above will produce optimal results and prevent unnecessary complications.

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14 Operation for Giant Duodenal Ulcer

Michael S. Nussbaum and Keyur Chavda

Introduction

Giant duodenal ulcer (GDU) is a variant of peptic ulcer disease and is defined as a duodenal ulcer crater that is benign and at least 2 cm in diameter. This subset of duodenal ulcers have historically resulted in greater morbidity than usual duodenal ulcers since, by definition, they involve the full thickness of the duodenal wall and occupy a significant portion of the duodenal bulb. Brdiczka first called attention to this entity in 1931 and emphasized the difficulty in diagnosing them with barium roentgenogram. Although once thought to be a rare variant, with the advent of flexible endoscopy and awareness of this entity numerous case reports and case series have been published in the literature. GDUs comprise approximately 1% to 2% of all duodenal ulcers and 5% of peptic ulcers requiring surgical intervention. In the initial reports, few patients were successfully treated with medical therapy. Surgery was favored as the treatment of choice for this disease, and high mortality rates were reported despite surgical intervention. Improvements in surgical techniques have revolutionized the operative treatment and outcome of this condition. However, today, with the widespread use of endoscopy, the introduction of histamine-2 receptor antagonists and later proton pump inhibitors (PPIs), medical treatment has replaced operation as the first line of treatment for the patient with GDU. Because of the large, penetrating nature of these ulcers, when not recognized and treated promptly, complications such as hemorrhage and perforation remain common, and GDUs are still associated with high rates of morbidity and mortality. Thus, surgical evaluation of a patient with GDU should remain an integral part of patient care in all cases. There are important differences when comparing GDUs to classic peptic ulcers, and they must be approached differently than their more common counterpart.

Standard-sized ulcer disease affects males greater than females at a rate of approximately 2 to 1, whereas it is 3 to 1 for GDU. The etiology of standard-sized and GDUs has been associated with two major contributing causes: recent usage of nonsteroidal anti-inflammatory drugs (NSAIDs) and *Helicobacter pylori* infection. However, the percentage of GDUs caused by *H. pylori* is less when compared to standard-sized ulcers, and NSAID use plays a more prominent role.

The most common presenting symptom is abdominal pain. Most patients describe the pain as involving the epigastric region, and some experience involvement of the right hypochondrium and/or radiation into the back, particularly when the ulcer penetrates into the pancreas. The pain is more intense and persistent than the pain found with usual ulcer patients. The pain is generally not relieved with food or antacids and weight loss is a frequent comorbidity.

Diagnosis

The most common emergency presentation of GDU is hemorrhage, which may manifest as melena, hematochezia, hematemesis, or any combination of the above, associated with anemia. The size of the ulcer and the surrounding inflammation may cause gastric outlet obstruction with nausea, vomiting, and weight loss. An inflammatory mass in the upper abdomen with associated weight loss, cachexia, malnutrition, and chronic abdominal pain can frequently mislead the clinician to suspect malignancy as the most likely diagnosis. Other important historical features include a past history of ulcer disease and the recent use of NSAIDs.

The size of the ulcer often causes replacement of the duodenal bulb, and as a result GDUs may be missed or misinterpreted as a deformed bulb, diverticulum, or pseudodiverticulum during a barium upper GI series. The advent and widespread use of endoscopy has markedly improved the ability to detect GDUs with greater accuracy. It is not uncommon to encounter a GDU during an endoscopy without the expectation of finding one, and it is important to measure the ulcer so as not to misdiagnose it as a simple peptic ulcer. The ulcers usually are quite deep and involve over 50% of the mucosal circumference of the duodenal bulb. It is essential to exclude a neoplastic source as the cause of ulcer formation with biopsy in the setting of GDUs, particularly when there is nodularity at the edge. A recent review of 52 cases of duodenal ulcers larger than 2 cm found a malignancy rate of approximately 19% (primary duodenal carcinoma in 15%, lymphoma and tuberculosis in 2% each).

Management

Medical Treatment

Prior to the introduction of histamine-2 receptor antagonists in the late 1970s, GDUs were managed primarily surgically. Before 1982, there were very few published reports of long-term successful medical management of GDUs. With the advent of new acid suppression medication, the discovery of *H. pylori* and its role in ulcer formation with the importance of eradication therapy, successful medical management of most GDUs is now possible. The most recent studies have demonstrated that PPI therapy is a safe and effective first-line treatment in stable patients and should decrease the eventual need for operative intervention. Thus, attempts at medical treatment of GDUs should consist of PPIs. Discontinuation of NSAIDs and antimicrobial treatment of *H. pylori* infection in conjunction with PPI therapy are important treatment adjuncts in the presence of these risk factors.

Surgical Treatment

Indications

Despite the marked improvement in outcome with medical therapy, GDUs are still associated with high rates of morbidity, mortality, and complications. All patients diagnosed with GDU should be evaluated promptly by a surgeon. Operation is indicated in patients with acute complications of GDU such as hemorrhage and perforation.

Intractability, recurrent disease, incomplete healing despite proper medical therapy, and gastric outlet obstruction may require surgical intervention. GDU with adherent clot or a visible vessel on index esophagogastroduodenoscopy (EGD) is a marker of an ulcer that is more likely to require early surgical intervention. Uncontrolled hemorrhage and perforation are the most common emergent indications for operation. Unresolving obstruction, intractable or recurrent bleeding, and fistula formation are some of the elective indications for operation. The chronic inflammatory changes associated with these conditions often make the operations in these patients technically quite challenging.

Preoperative Assessment

A detailed clinical history and physical examination is required before elective surgery. Radiological study with contrast (upper gastrointestinal series) is useful in the evaluation of the duodenum and is useful in defining the anatomy of the upper digestive tract. EGD is useful in establishing the diagnosis of GDU and identifying features that may preclude successful medical treatment. EGD is also useful for monitoring the healing of the ulcer after medical therapy in cases of intractability or recurrent disease. Other conditions which may mimic GDU are pseudodiverticulum of the bulb or true diverticulum of the postbulbar area and a neoplasm such as carcinoma or lymphoma of the duodenal bulb. These conditions when suspected must be ruled out prior to surgery. If there is a suspicion of neoplasm, EGD is required for examination and biopsy to rule out malignancy.

The nutritional and immune status of the patient should be assessed prior to operation. Patients with chronic ulcer disease and gastric outlet obstruction are more prone to malnutrition with a higher incidence of perioperative complications. Severely malnourished patients may benefit from a course of preoperative enteral or parenteral nutrition. Patients with significant medical comorbidities will require thorough evaluation and optimization of their medical conditions prior to elective operation.

Operation

A definitive acid-reducing operation is the procedure of choice for patients with GDU requiring surgical intervention. Truncal vagotomy and antrectomy with removal of the involved duodenum is the procedure of choice whenever possible. When the inflammation and edema of the duodenum is not a factor, a Billroth I reconstruction can occasionally be performed (see Chapter 1 for truncal vagotomy, antrectomy, and Billroth I). Frequently, however, due to the nature of GDU, the involved duodenum is so inflamed and indurated that duodenal dissection and anastomosis would be hazardous. Thus, a Billroth II gastrojejunostomy is the most frequent method of reconstruction utilized in these patients (see Chapter 3 for truncal vagotomy, antrectomy, and Billroth II).

Because of the large size of the ulcer bed and its proximity to such vital structures as the pancreas, common bile duct, and their blood supply, it may be best to leave the ulcer bed in situ, resecting the duodenum around the ulcer bed and closing the duodenal stump. Duodenal stump leak is a major source of morbidity and mortality after resection of duodenum in these patients. Numerous methods and modifications of the standard duodenal stump closure have been described in an effort to prevent leakage in this setting. The use of a duodenostomy tube is a safe, rapid, and effective means of managing the difficult duodenal stump so often encountered in this setting. Tube duodenostomy involves insertion of a tube through the second portion of the duodenum to encourage formation of a controlled duodenocutaneous fistula. Similarly, a retrograde tube can be threaded through the wall of the jejunum downstream at the site of the placement of antegrade feeding jejunostomy (*drain-me feed-me tubes*) to provide decompression of this portion of the afferent limb.

The Nissen closure of the duodenal stump can be used to reinforce a difficult stump. After the duodenum is transected, the open lumen is sutured to the capsule of the pancreas (Fig. 14.1A). Another technique is the Bancroft procedure and its various modifications which entail performing an antrectomy but leaving the distal 3 to 4 cm of antrum proximal to the pylorus and performing a mucosectomy in which the mucosal layer of the antral stump and the pylorus are dissected away from the submucosa and

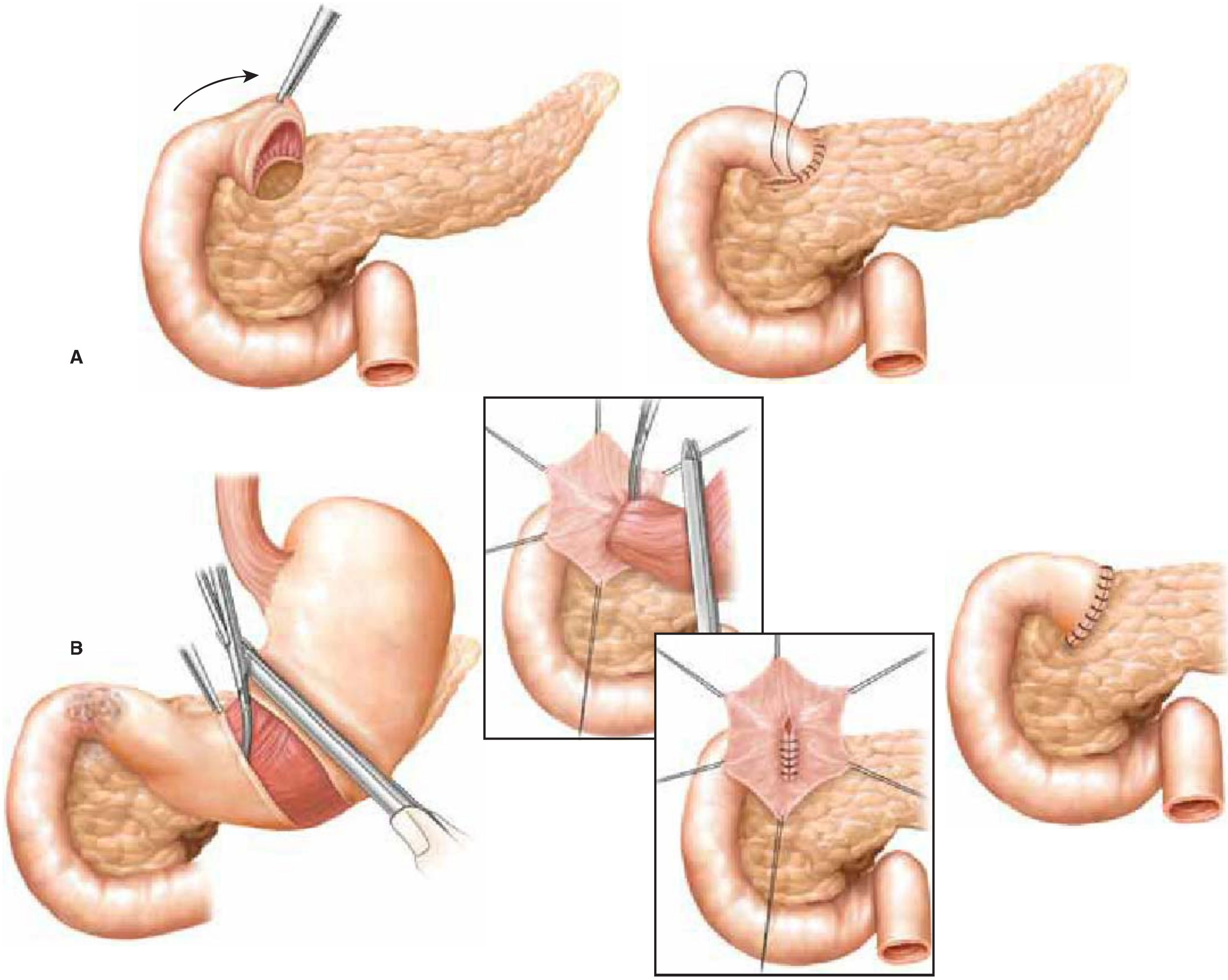

Figure 14.1 (**A**) Nissen closure. This method, often employed when the duodenum is scarred to the pancreatic capsule, is performed by first transecting the duodenum. The duodenal stump is then sutured to the pancreatic capsule or duodenal wall left in place on the pancreatic capsule. (**B**) Bancroft closure. In this method of duodenal stump closure, the stomach is transected 3 to 4 cm proximal to the pylorus, where tissue is less fibrotic. The antral mucosa in the duodenal stump is then dissected away from the submucosa beyond the pylorus and into the duodenum. This is secured with a purse-string suture, and the seromuscular layer is closed over the stump.

removed. A purse-string suture is then used to close the pylorus from inside; then the submucosa and the muscularis layers of the prepyloric stomach are used to reinforce the closure of the duodenal stump (Fig. 14.1B). An important principle is preservation of the distal antral blood supply because necrosis of the stump is a risk with this procedure. Either of these closure techniques can be combined with duodenostomy tube decompression.

When emergency surgery is required for continuing hemorrhage from a duodenal ulcer, it is common practice to oversew the bleeding ulcer and at the same time perform a vagotomy and pyloroplasty. But in the case of a GDU, where the blood may be coming directly from the gastroduodenal or pancreaticoduodenal artery, this approach is insufficient and carries a high risk of recurrent bleeding. Truncal vagotomy and either pyloroplasty or gastrojejunostomy should be avoided. The ulcer must be separated from the duodenum and the blood supply to the ulcer must be appropriately ligated.

Laparoscopic approaches for gastroduodenal disease are well established. The minimally invasive approach allows for the possible benefits of less pain, less wound complications, and shorter hospital stay (see Chapter 2 for laparoscopic vagotomy, antrectomy, and Billroth I and Chapter 4 for laparoscopic vagotomy, antrectomy, and Billroth II). However, because of chronic inflammation around the ulcer area and surrounding organs, the laparoscopic approach is sometime difficult when treating GDU. The surgeon should be prepared to convert to an open approach if duodenal inflammation makes it difficult to perform a safe and expeditious resection.

CONCLUSIONS

Historically, the two key features of GDU disease were the difficulties in prompt diagnosis and the failures of medical management. The advent and widespread use of endoscopy has made prompt diagnosis of GDU more accurate and easy to obtain. Once a diagnosis has been made, initiation of therapy can begin. Uncontrolled hemorrhage, perforation, and unstable patients should undergo prompt operative management with vagotomy and antrectomy. Stable patients may be safely treated initially with medication. However, close observation and repeat endoscopic evaluation is essential in the successful medical management of these patients. Furthermore, a malignant etiology may be more common than previously suspected, and liberal use of endoscopic biopsies are warranted in the setting of GDU, particularly those with nodular-appearing edges. Due to evolving endoscopic and medical therapies, the management of GDUs has changed. What was once a disease that was difficult to diagnose and managed solely with surgical intervention has become one easily diagnosed and potentially treated medically. It is of utmost importance that physicians recognize GDUs as being different than their standard-sized counterparts and that we continue to further our understanding of this entity.

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15 Distal Subtotal Gastrectomy and D1 Resection

K. Roggin and Mitchell C. Posner

INDICATIONS/CONTRAINDICATIONS

Gastric adenocarcinoma is the fourth most common malignancy and the second leading cause of death worldwide annually. In the United States, it remains a relatively rare cancer (in 2010, approximately 22,000 new cases were diagnosed). Gastric carcinoma is associated with chronic exposures to environmental carcinogens (*Helicobacter pylori*, nitrosamines, tobacco exposure), inflammatory conditions of the stomach (atrophic gastritis), pernicious anemia, blood type A, and less commonly, genetic mutations in the E-cadherin gene, *CDH-1* (hereditary diffuse gastric cancer). The relatively poor overall survival reported in most Western series is likely influenced by the high percentage of patients who are diagnosed with locally advanced or metastatic disease. Symptoms remain nonspecific early in the course of the disease, but advanced tumors can cause GI hemorrhage, gastric outlet obstruction, and profound weight loss. Curative treatment requires an appropriate gastrectomy with adequate regional lymphadenectomy. Peri- or postoperative treatment with chemotherapy and/or radiation has been shown to reduce recurrence and improve survival. Tumor location, clinical stage of disease, and patient performance status influence the decision to perform a subtotal versus total gastrectomy. Lymphatic metastases should be treated with an appropriate lymphadenectomy for optimal locoregional control of disease. Early-stage cancers may be cured by complete endoscopic mucosal resection (e.g., T1a cancers) or surgical gastrectomy; chemotherapy and/or radiation are effective adjuvant treatments that have been associated with improved disease-free and overall survival in stage II and III disease. Optimal management should be individualized with the input of a multidisciplinary tumor board. Two treatment paradigms are accepted as standards of care:

- Perioperative chemotherapy using EOX (Epirubicin–Oxaliplatin–Capecitabine) or equivalent chemotherapy for three cycles (nine weeks) before and after radical gastrectomy (MAGIC trial regimen).
- Subtotal or total gastrectomy followed by chemotherapy and external beam radiation (McDonald regimen, SWOG 0116).

PREOPERATIVE PLANNING

NCCN guidelines (www.nccn.org) suggest a comprehensive staging workup for patients with resectable gastric cancer.

- Comprehensive history and physical examination
- Complete upper endoscopy defining the location of the tumor within the stomach, extent of intragastric spread, and relationship of the cancer to the gastroesophageal junction.
  - All biopsies should be reviewed by a dedicated GI pathologist to determine the histologic subtype (intestinal, Lauren's diffuse type, signet ring cell adenocarcinoma, adenosquamous carcinoma) and degree of differentiation.
- Endoscopic ultrasonography should be considered in select patients who are candidates for neoadjuvant chemotherapy protocols. This modality is accurate at assessing the depth (T stage) of invasion, presence of metastatic regional lymphadenopathy, and distant metastatic disease to the liver or peritoneal cavity (liver metastases, peritoneal implants, and/or ascites).
- Contrast-enhanced, triphasic (pre-, arterial-weighted, and portovenous phases) multirow detector computed tomography of the chest, abdomen, and pelvis.
- Positron-emission tomography (PET scans) remains an experimental diagnostic staging modality in gastric adenocarcinomas, as only two-thirds of these mucin-producing or signet ring adenocarcinomas have the ability to concentrate the radiotracer fluorodeoxyglucose. PET–CT fusion scanning has been reported to improve the diagnostic accuracy compared with either CT or PET scans alone.
- Staging laparoscopy ± peritoneal washings (cytology) should be considered in patients with ≥T3 or node-positive cancers; as many as 20 to 30% of patients with negative radiographic and endoscopic imaging will have occult M1 or stage IV disease on laparoscopy.
  - Peritoneal cytology appears to be an independent predictor associated with death from gastric adenocarcinoma.

SURGERY

Complete surgical resection of gastric cancers is the only treatment modality associated with long-term survival. The tumor stage, location, and performance status of the patient influence the optimal type of resection (subtotal vs. total gastrectomy). Two randomized prospective trials have shown that the estimated overall survival after distal subtotal gastrectomy is equivalent to total gastrectomy for distal gastric cancers. Total gastrectomy is associated with higher postoperative complication rates, more frequent concomitant splenectomy, and longer inpatient length of stay. In addition, it is often associated with significant long-term protein-calorie malnutrition and functional impairment. Proximal gastrectomy has not been rigorously compared with total gastrectomy, but it offers a reasonable alternative to total gastrectomy for cancers of the cardia and gastroesophageal junction (GEJ). In general, this procedure has a higher frequency of recalcitrant postoperative biliary reflux. Radical gastrectomy requires a comprehensive understanding of the arterial supply of the stomach and duodenum (Fig. 15.1), lymphatic drainage basins, and physiologic consequences of decreasing the volume of the gastric reservoir. The optimal extent of regional lymphadenectomy remains controversial. Two landmark-randomized controlled trials failed to show a short-term survival benefit with D2 lymphadenectomy. Results from both a recent trial by Wu et al. and a 15-year re-analysis of the Dutch gastric cancer trial suggest a small absolute survival benefit for patients who were treated with extended lymphadenectomy (>D1).

Operative principles include the following:

- Complete laparoscopic and open assessment of occult, sub-radiographic metastases to the liver, peritoneal cavity, adrenal glands, and distant lymphatic basins.

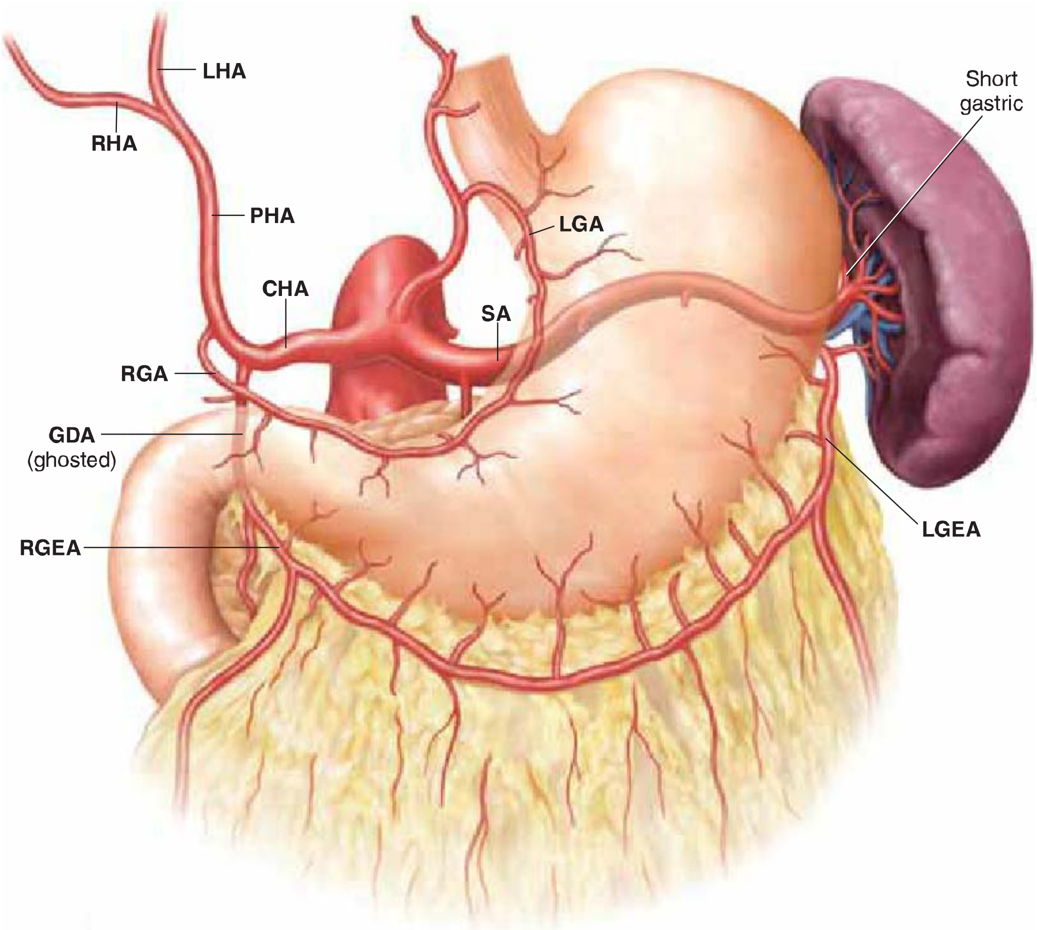


Figure 15.1 Arterial supply to stomach and duodenum. LGA, left gastric artery; SA, splenic artery; CHA, common hepatic artery; PHA, proper hepatic artery; LHA, left hepatic artery; RHA, right hepatic artery; RGA, right gastric artery; GDA, gastroduodenal artery; RGEA, right gastroepiploic artery; lGEA, gastroepiploic artery.

- Complete resection of the primary tumor with at least 5-cm proximal and distal margins.
- Appropriate regional lymphadenectomy as indicated by the location of the primary tumor (proximal, middle, and distal stomach) and stage of disease. In the absence of clinical lymphadenopathy in the D2 drainage basins (celiac axis distribution), a complete D1 lymphadenectomy may be sufficient treatment. Resecting at least 15 lymph nodes appears to ensure adequate staging accuracy.
  - D1 lymphadenectomy involves removal of the perigastric lymph nodes along the lesser curvature (stations 1, 3, and 5) and greater curvature (stations 2, 4, and 6) (Fig. 15.2).
  - D2 lymphadenectomy extends the lymphatic sampling to all of the lymph nodes around the celiac axis and its named vessels (stations 7–11).
- Gross and histologic intraoperative margin assessment.
- Reconstruction of GI tract continuity with appropriate conduit to maximize function and reduce the incidence of postgastrectomy syndromes.

Operative Positioning and Setup

Patients are positioned on the operating room table in the supine position with appropriate padding. Sequential compression devices and a single dose of 5,000 units of subcutaneous heparin are administered prior to induction of anesthesia to reduce the incidence of perioperative thromboembolic events. An oro- or nasogastric tube and Foley catheter are placed into their respective locations after the patient has been sedated and successfully intubated. Complete pharmacologic neuromuscular blockade is essential to maximize operative exposure and minimize incision length. Broad-spectrum antibiotics covering gastric flora (e.g., second-generation cephalosporins) are given intravenously

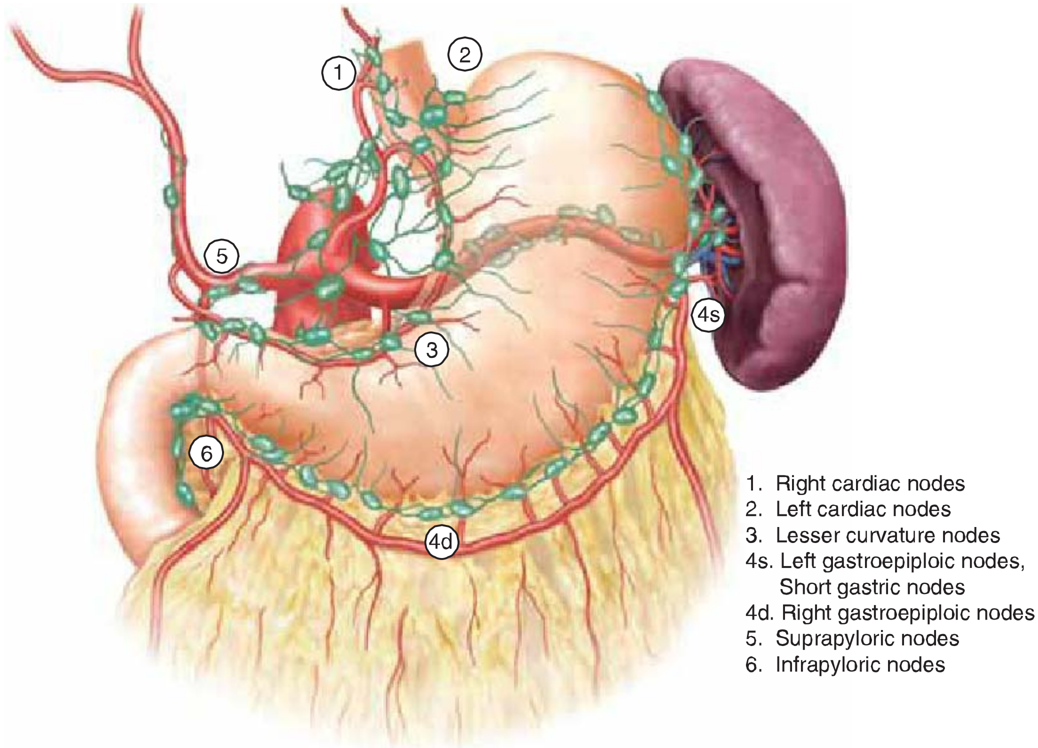


Figure 15.2 D1 lymphadenectomy stations.

within 1 hour prior to the incision; antibiotics are routinely re-dosed within one-half life (generally 4 to 6 hours) if needed. The patient's left arm is usually padded and tucked to facilitate the placement of the retractor arm. The skin is prepared with a chlorhexidine solution and allowed several minutes to completely dry before draping the patient. An antibiotic-impregnated impermeable barrier is placed on the abdominal skin.

Operative Technique

Diagnostic Laparoscopy

All patients with gastric cancer operation require complete operative staging prior to resection. A formal diagnostic laparoscopy is performed to rule out radiologically occult metastatic disease. Generally, two trocars are used: a 12-mm supraumbilical incision for the Hasson port and a 5-mm port in the subcostal position along the left midclavicular line. The peritoneal cavity should be systematically explored to identify intrahepatic metastases, peritoneal carcinomatosis, drop metastases (i.e., Krukenberg tumors), or local extension into surrounding viscera (e.g., linitis plastica). If peritoneal washings are required, the laparoscopy is usually performed as a staged procedure. In general, the results of the peritoneal washings take 24 to 48 hours to be completed. Approximately 50 to 100 mL of normal saline is instilled into the left upper quadrant, right upper quadrant, and pelvis; 30 mL of fluid from each location is collected and sent in separate containers for cytology evaluation.

Exploratory Laparotomy

The peritoneal cavity is thoroughly explored through an upper midline (preferred) or bilateral subcostal incision to confirm the absence of metastatic disease. We prefer an Omni or Thompson retractor to separate the wound edges for optimal exposure. The falciform ligament should be divided and the liver thoroughly evaluated by palpation and if needed, hand-held ultrasonography. The entire small bowel and abdominal viscera are examined.

Distal Subtotal Gastrectomy Resection of the Stomach

- The nasogastric tube is advanced into the duodenal bulb and used as a "handle" to elevate the gastric body. The transverse colon is retracted in a caudal direction to facilitate mobilization of the greater omentum off the transverse colon with the electrosurgery device. The micronodular, granular appearance of the greater omentum is reflected in a cranial direction and the homogenous, light yellow fat of the transverse mesocolon is retracted caudally and preserved. The dissection in the avascular plane is extended posteriorly down to the inferior border of the pancreas. If possible, you should remove the visceral peritoneum off the anterior surface of the body and tail of the pancreas. The soft tissue dissection continues to the celiac axis. If a D2 lymphadenectomy is performed, the lymphatic stations (7–11) around the celiac-named vessels will be harvested.
- The duodenal bulb is mobilized off the retroperitoneum and hepatic flexure colonic mesentery. It is critical to identify and preserve the suprapancreatic portion of the extrahepatic bile duct. The gastroduodenal artery is identified and preserved.
- The omental mobilization continues laterally to the hepatic and splenic flexures of the colon. The lateral borders of the omentum are divided using a tissue transection device (i.e., Ligasure or Enseal). On the right side, special care is taken to avoid injuring the duodenal sweep and head of pancreas. On the left side, the omental resection is facilitated by dividing the white line of Toldt to ensure that the splenic flexure of the colon is fully mobilized and not inadvertently injured. The omentum is divided along the splenic hilum with great care taken to avoid capsular traction injuries to the spleen or splenic artery/vein.
- The short gastric vessels are divided up to the level of the gastric fundus using the tissue transection device.
- The right gastroepiploic artery and vein are divided at the level of the superior mesenteric vein between 2-0 silk ties; the proximal end is usually reinforced with a 4-0 prolene suture ligature.
- All lymphatic tissue along the gastroepiploic vessels is reflected toward the planned gastric specimen.
- The supra- and infraduodenal lymph nodes are included in the D1 dissection.
- The duodenal bulb is circumferentially skeletonized by dividing all small feeding vessels between 3-0 silk ties.
- The duodenum is divided with a linear mechanical stapling device with either a 2.5-mm or 3.5-mm load. Hemostasis is achieved along the staple line with the electrosurgery device. Although not necessary, the staple line can be inverted with interrupted 3-0 silk or 3-0 Maxon sutures.
- The lesser (gastrohepatic) omentum is divided sharply from medial to lateral after ruling out the presence of a replaced left hepatic artery. The right gastric artery is divided close to its origin off the common/proper hepatic artery; the proximal end is usually reinforced with a 4-0 prolene suture ligature.
- The lymphatic tissue along the lesser curvature of the stomach is reflected toward the planned specimen. The left gastric artery and vein are divided at its origin (usually along the superior border of the pancreas) between Kelly clamps, ligated with 2-0 silk ties, and the proximal end is reinforced with a 4-0 prolene suture ligature. The left gastric artery is identified because it tends to have a short (posterior to anterior) course before curving laterally along the proximal lesser curvature of the stomach.
- At this point, the proximal body of the stomach should be completely skeletonized and prepared for transection.
- It can be helpful to perform an intraoperative upper endoscopy to confirm the precise proximal extent of the tumor prior to resection. We usually have our assistant mark this location on the serosa of the stomach with an identifiable silk stitch. The stomach is decompressed and the nasogastric tube, endoscope, and esophageal temperature probe are completely removed.
- The stomach is transected at least 5 cm proximal to the superior extent of the tumor. The stomach can be divided with a 90-mm TA™ stapler (Covidien Surgical, Mansfield,

MA), sequential fires of the Endo GIA™ Ultra Universal reticulating stapler (Covidien) with Tri-Staple™ technology (purple medium/thick loads with 3.0-mm, 3.5-mm, 4.0-mm staples). If the TA stapler is used, the distal transected end of the stomach should be occluded with a long atraumatic bowel clamp to avoid spillage. Hemostasis is achieved along the staple line with the electrosurgery device. Additional topical hemostasis along the staple line can be controlled with interrupted 3-0 silk horizontal mattress sutures under the staple line. The gastric staple line is often inverted with 3-0 silk sutures in Lembert fashion. We keep the corner sutures attached to small Kelly clamps for traction during the construction of the anastomosis.

- The specimen should be transferred to a back table for orientation. We generally mark the proximal margin with an easily identifiable suture. The specimen is hand-delivered to pathology to have all the margins inked and the specimen completely opened. If there is at least a 5-cm gross margin proximal and distal to the mass and no evidence of linitis plastica, frozen sections are not routinely performed. If there is concern about either margin, it is reasonable to request a frozen section of the proximal or distal margin. It should be noted that most pathologists cannot assess the entire circumferential margin and usually sample random portions of the margin closest to the tumor.

Distal Subtotal Gastrectomy Reconstruction

- The peritoneal cavity is copiously irrigated with normal saline.
- Reconstruction of the GI tract (Fig. 15.3) can either be accomplished with a
  - Billroth I gastroduodenostomy
  - Billroth II gastrojejunostomy ± enteroenterostomy (Omega loop)
  - Roux-en-Y gastrojejunostomy

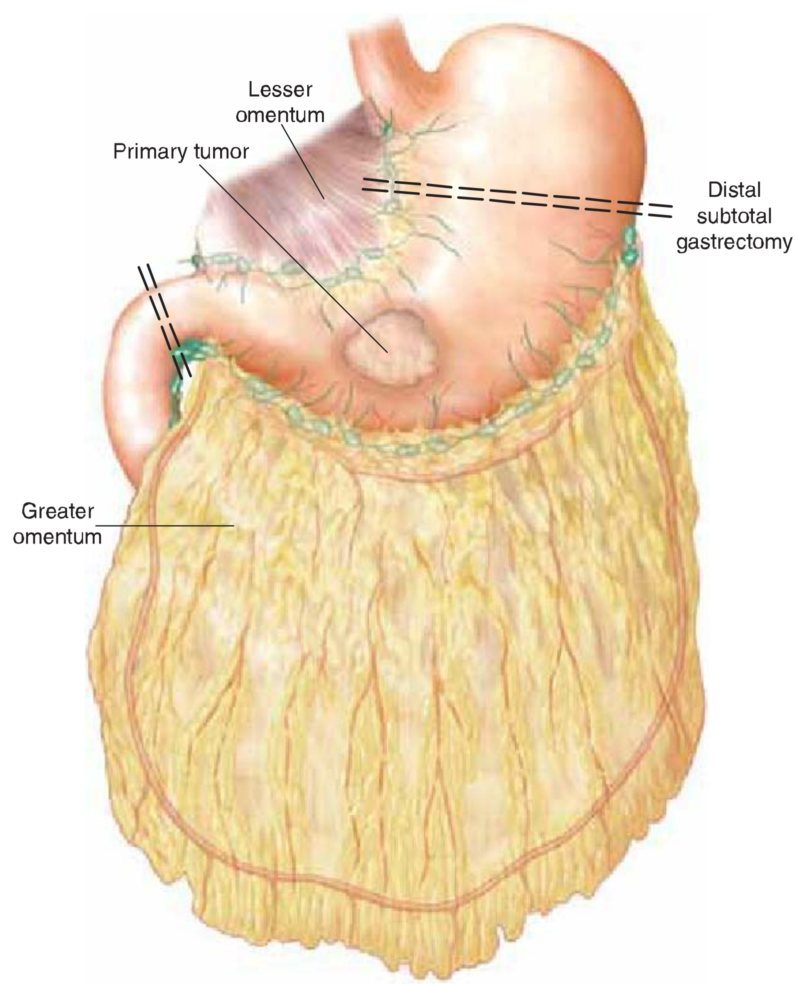


Figure 15.3 Reconstruction of the gastric remnant.

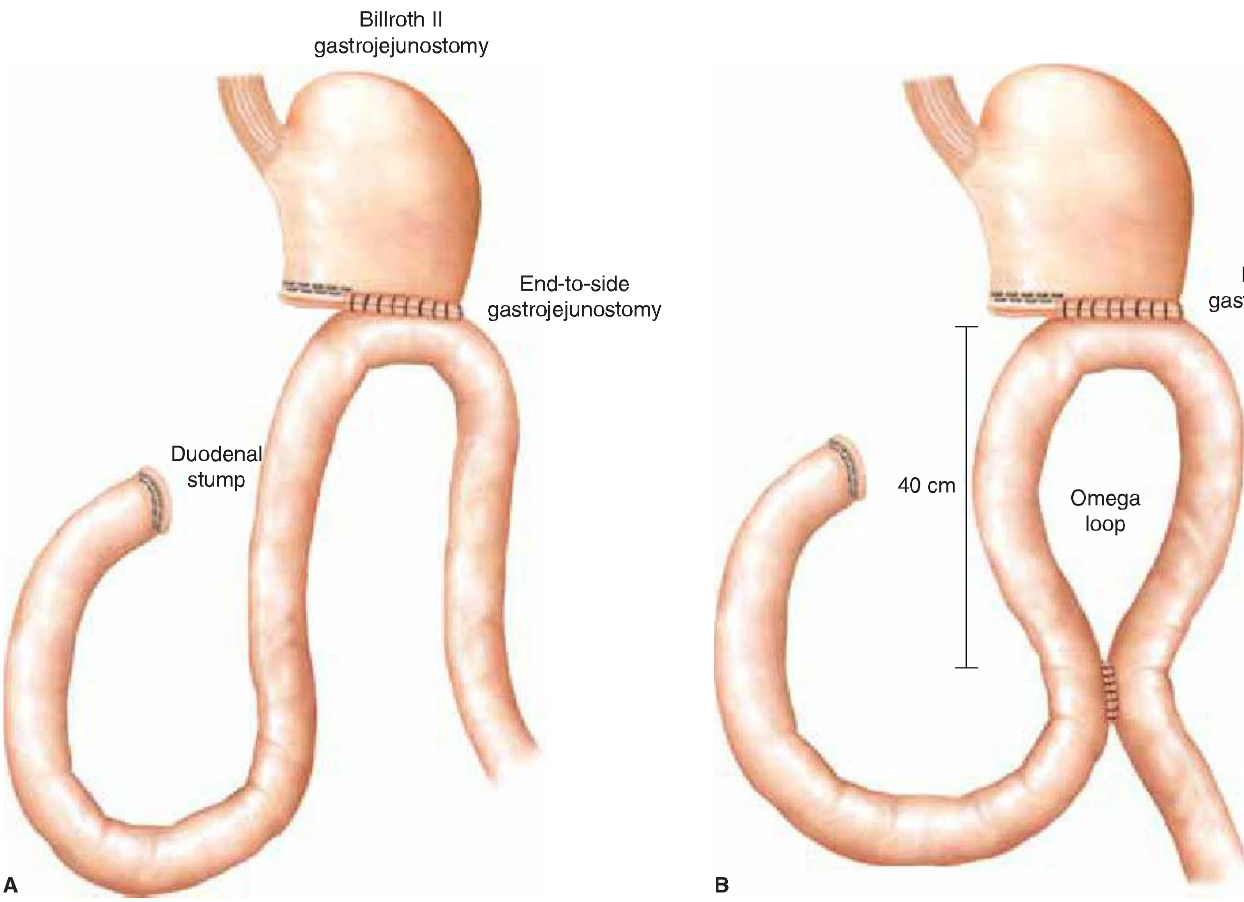
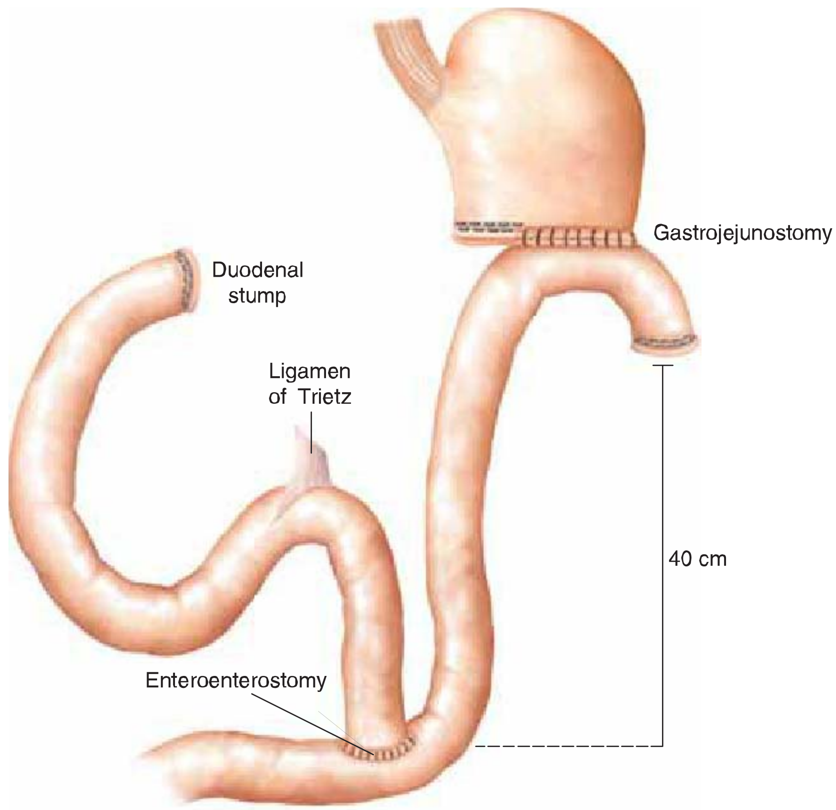


Figure 15.4 Billroth II gastrojejunostomy (**A**) and with an enteroenterostomy to prevent alkaline reflux (**B**).

- The choice of reconstruction depends on the likelihood of recurrence, body habitus, flexibility of the small bowel mesentery, estimated recurrence-free survival, and life expectancy of the patients. In general, we prefer to use the Roux-en-Y gastrojejunostomy.
  - A Billroth II gastrojejunostomy (Fig. 15.4A), ± an Omega loop (40 cm enteroenterostomy to prevent alkaline reflux, Fig. 15.4B) can be completed using a hand-sewn technique or with mechanical stapling devices. We use either interrupted 3-0 silk or continuous Maxon (monofilament polyglyconate synthetic absorbable sutures, Covidien) sutures for both anastomoses. The mucosal reapproximation is performed in Connell fashion. The gastrojejunostomy is typically performed along the posterior wall of the stomach (1 to 2 cm from the staple line) to facilitate drainage of the gastric remnant. A stapled anastomosis is constructed by aligning the sides of the stomach and jejunum with 3-0 silk sutures at least 1 cm away from the gastric staple line. A small gastrotomy and antimesenteric jejunotomy are created to accommodate an endo- or open Endo-GIA™ stapler (3.5-mm or 3.8-mm/blue loads). A generous common channel is created and special care is taken to inspect and control bleeding from the staple line. The common enterotomy is aligned transversely with Allis clamps and closed with a 45-mm or 60-mm TA™ stapler (4.8-mm/green load); alternatively, this can be closed in two layers with sutures.
  - Roux-en-Y gastrojejunostomy (Fig. 15.5) is the best method of reducing the long-term complications that have been associated with Billroth II reconstructions (i.e., marginal ulceration, alkaline reflux gastritis, afferent loop syndrome, etc.). This may be more technically challenging in short/obese patients with a foreshortened or tethered small bowel mesentery. The jejunum is divided approximately 40 cm from the ligament of Treitz with an Endo-GIA™ stapler (3.5-mm/blue load). A window is created in the transverse mesocolon to the left of the middle colic vessels. The roux limb should be delivered without tension and with proper mesenteric orientation. The hand-sewn gastrojejunostomy is created along the posterior wall of the stomach

Figure 15.5 Roux-en-Y gastrojejunostomy.



using either 3-0 silk or Maxon sutures in two layers; in general, we try to construct the anastomosis along the lateral aspect of the gastric remnant to avoid the "angle of sorrow" along the lesser curvature near the GEJ. The mesocolic defect is closed with 3-0 polysorb sutures in interrupted fashion; special care should be taken to avoid narrowing the aperture leading to obstruction of the gastric limb. A side-to-side enteroenterostomy is created in a similar fashion so that the total length of the roux limb (distance from the gastrojejunostomy to enteroenterostomy) is at least 35 to 40 cm in length. This is performed to prevent biliary reflux into the gastric remnant. This technique can be performed using either hand-sewn or stapled anastomoses.

- Billroth I gastroduodenostomy has excellent long-term functional results by preserving the passage of food through the native duodenum. This appears to preserve the proper physiologic balance of GI endocrine secretion (pancreatic exocrine enzyme secretion, bile flow) and may contribute to the inhibition of gastric acid secretion through the production of somatostatin and other hormones. The reconstruction is typically performed in an end-to-end fashion and can be performed using an EEA™-type mechanical stapling device. This technique should be avoided in cancers that are likely to recur locally (i.e., diffuse type, linitis plastica, or node-positive primary tumors.)

- The anastomosis is visually inspected for integrity. A repeat endoscopy can be performed to visualize the anastomosis for hemorrhage. The distal bowel is occluded with an angled atraumatic bowel clamp prior to insufflation. The patient is placed into a steep Trendelenburg position and the upper abdomen is filled with warm normal saline. An air leak test is performed with maximum distention of the stomach. If no leak is identified, the scope is withdrawn after the stomach has been completely deflated.

Conclusion of Surgical Procedure

- The peritoneal cavity is copiously irrigated with normal saline lavage and antibiotic irrigation (e.g., bacitracin).

- Meticulous hemostasis is achieved, especially along the retroperitoneal dissection planes and on all named arterial transection sites.
- Nasogastric tubes are generally not replaced.
- Feeding jejunostomy tubes are not routinely placed unless the patient is frail or severely malnourished.
- After the preliminary sponge, needle, and instrument counts are verified, the anterior abdominal wall fascia is closed in a single layer using an absorbable monofilament suture (i.e., #0 or #1 looped Maxon).
- Meticulous hemostasis is obtained in the subcutaneous tissues with the electrosurgery device. The subcutaneous tissues are irrigated with bacitracin solution.
- The skin is closed with staples and the wound is covered with a water-repellant dressing.

POSTOPERATIVE MANAGEMENT

- Most patients can be safely managed in a multispecialty bed unit. We only admit to the intensive care unit if the patient has significant medical comorbidities, excessive intraoperative blood loss/case length, or if there were respiratory difficulties during the case.
- Gum and hard candy are encouraged as an economical method of stimulating bowel activity.
- In general, we do not restrict oral intake until the patient passes flatus or has a bowel movement. Instead, we individualize the decision to advance oral intake based on the clinical status of the patient. Aspiration pneumonia is a serious adverse event and should be vigilantly prevented by minimizing narcotics, keeping the head of bed elevated at 30 degrees, and examining the patient frequently for distention that could indicate ileus or obstruction.
- Pain is controlled via patient-controlled analgesic devices (e.g., PCA). We encourage patients to limit narcotics to prevent acute delirium and ileus. Ketorolac can be used for a limited duration (generally 4 to 6 doses) to relieve the musculoskeletal pain associated with the incision and facilitate dose reductions in narcotic analgesics. Since these agents potentiate anticoagulants, we do not concomitantly administer blood thinners during the time that ketorolac is being used. This medication should be used with caution or at a reduced dosage in elderly patients to prevent acute renal failure.
- DVT prophylaxis begins in the OR (subcutaneous heparin and sequential compression devices) and continues postoperatively. Our practice has been to give prophylactic enoxaparin for 28 days postoperatively to reduce the risk of life-threatening thromboembolic events.
- Routine upper GI contrast studies are not routinely performed unless clinical symptoms suggest an anastomotic leak.

COMPLICATIONS

Complications following distal subtotal gastrectomy include hemorrhage, surgical site infection (superficial or organ space), deep venous thrombosis, aspiration/nosocomial pneumonia, and cardiac events. Rare complications specific to the procedure include

- Anastomotic leak
- Afferent loop syndrome
- Duodenal stump leak
- Delayed gastric emptying
- Alkaline reflux gastritis
- Peptic ulcer disease
- Marginal ulceration

Long-term complications include adhesive small bowel obstruction, ventral hernia, internal small bowel hernia (Peterson's hernia with Roux-en-Y gastrojejunostomy), and anastomotic stricture.

RESULTS

The 5-year overall survival rate of all patients with gastric cancer is approximately 15 to 20%. If the disease is confined to the stomach, complete resection is associated with long-term survival rates >50%. Independent predictors of survival after complete R0 resection include age, sex, primary site, histologic subtype (e.g., diffuse, mixed, intestinal), number of positive lymph nodes, and depth of primary tumor invasion.

CONCLUSIONS

The optimal oncologic treatment of distal and midgastric cancer is radical subtotal gastrectomy and regional lymphadenectomy. Extended lymphadenectomy (>D1 nodal resection) improves the accuracy of pathologic staging, increases postoperative morbidity, and may improve survival in a subset of patients with clinically positive nodal metastasis. Perioperative chemotherapy for locally advanced tumors has been shown to increase the rates of resectability and improve long-term survival.

Acknowledgments

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16 Laparoscopic Subtotal Gastrectomy and D1 Resection

Namir Katkhouda, Helen J. Sohn, John C. Lipham, and Joerg Zehetner

INDICATIONS/CONTRAINDICATIONS

Resectable gastric cancers involving the body and antrum are the main indication for subtotal gastrectomy with Roux-en-y gastrojejunostomy and regional lymphadenectomy (D1-resection: includes the perigastric lymph nodes stations 1–6 and the nodes along the left gastric artery station 7). A minimum of 5 cm proximal margin is required. Laparoscopic approach can be used when there are no contraindications for laparoscopy, such as multiple prior upper gastrointestinal operations, liver cirrhosis, poor cardiac status proven on stress test, pregnancy, or inability for the patient to give consent for laparoscopy.

Uncommon Indications

- Benign tumors of the stomach such as gastrointestinal stromal tumor located in the proximal stomach that is too large for a wedge resection. No lymphadenectomy is required.
- Complications of ulcer diseases such as bleeding and obstruction.
- Gastroparesis with uncontrollable vomiting. Near total gastrectomy can be performed after all medical and other therapies have been exhausted. No lymphadenectomy is required.
- Conversion of distal gastrectomy with gastroduodenostomy or loop gastrojejunostomy.

Contraindications for subtotal gastrectomy are cancers of the proximal stomach or diffuse gastric cancer both of which will necessitate a total gastrectomy for an R0 resection with disease-free microscopic margins. Metastatic cancers are treated by chemotherapy unless bleeding, obstruction, or perforation is present. Bleeding or perforated gastric cancers should be treated with a resection and obstructed cancers can be treated by either a resection or a bypass. Invasion of adjacent organs such as colon, liver, or spleen is not a contraindication to resection.

PREOPERATIVE PLANNING

- Upper endoscopy is performed to assess the location and the extent of the disease.
- Endoscopic ultrasound is performed to assess the lymph nodes in the celiac axis and obtain a fine needle biopsy for staging and planning treatment.
- CT scan of the chest, abdomen, and pelvis is important to assess the presence of metastasis in distant lymph nodes, peritoneum as studding and ascites, or distant organs.
- PET scan is used to further assess for metastatic diseases. There may be limited usefulness if the primary tumor does not show increased uptake of the tracer.
- Standard cardiopulmonary workup should be done to assess operability of the patient.

SURGERY

Laparoscopic surgery should be performed using the same principle as open surgery.

Positioning

Patient is supine in low lithotomy position to allow the surgeon to stand between the legs. Both arms should be tucked to allow placement of retractor posts onto the bed at the nipple level. Patient is placed in steep reverse Trendelenburg position to expose the upper abdominal structures with gravity. This is the ideal position for all upper gastrointestinal operations, and the surgeon faces the screen above the head of the patient. The first assistant is to the right and the camera assistant to the left of the patient. This enables the surgeon to have a straight view of the relevant screen. A Mayo stand for the surgeon's instruments is usually placed to the surgeon's right and the scrub technician stands to the left. It is important in this position for advanced upper abdominal procedures to avoid deep venous thrombosis by using bilateral lower extremity sequential compression devices and comfortably padding the knees, ankles, and legs.

Port Placement

Before starting the operation it is important to insert a nasogastric tube to decompress the stomach, thus avoiding any injury to the stomach upon insertion of the Veress needle.

The first 12-mm trocar is introduced after the creation of the pneumoperitoneum with the Veress needle using an optical port in the midline superior of the umbilicus (Fig. 16.1). The reason for midline rather than left paramedian for this port is to extend this incision later in the procedure for specimen removal and introduction of the circular stapler if used. This small incision can also be used for jejunojejunostomy and optional feeding jejunostomy tube placement. This port is used for the camera.

A second 12-mm trocar is placed in the left subcostal position in the midclavicular line as a primary working port.

Three additional 5 mm or 10 mm trocars are used:

- One in the subxyphoid position used to make a subcutaneous tract for the introduction of the Nathanson Hook Liver Retractor (Automated Medical Products Corporation, Sewaren, NJ). This retractor is secured to the right bed post using an Iron Intern (Automated Medical Products Corporation, Sewaren, NJ).
- Two retracting ports are placed in the right and left subcostal margins.

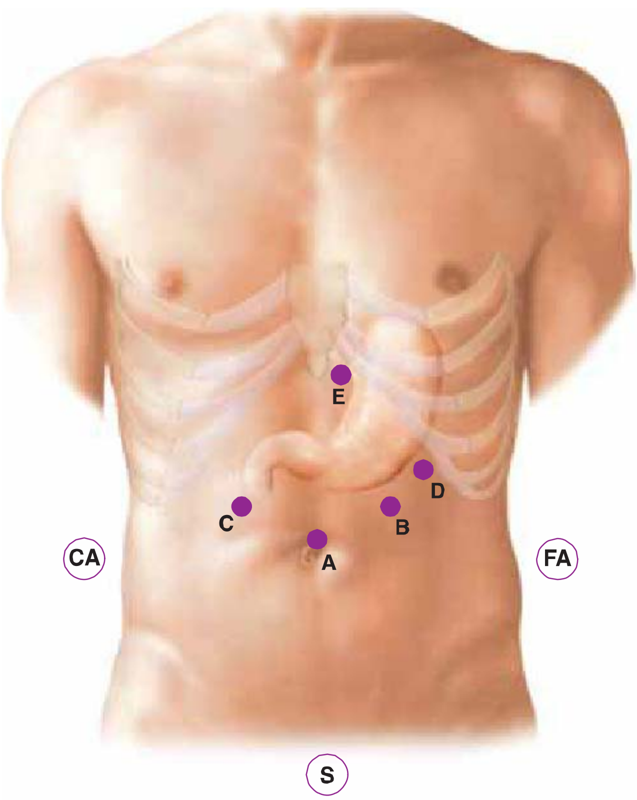


Figure 16.1 Port position for laparoscopic gastrectomy. A, umbilicus; B, surgeon's right hand (scissors); C, surgeon's left hand (grasper), D, assistant's grasper; and E, subxiphoid port. S, surgeon; FA, first assistant; and CA, camera assistant.

Technique

We will describe two common techniques:

Greater Curvature First Technique

Before the actual dissection starts, it is important to check on the various landmarks: the curvatures of the stomach, the gastrocolic ligament and the gastroepiploic arcade, the inferior aspect of the antrum, the duodenum, the pyloric muscle, the lesser sac, and the right gastric artery. The limit of the antrum and the proposed site of the gastrojejunostomy is marked using electrocautery. The assistant begins the procedure by retracting the greater curvature of the stomach using the lateral port. The surgeon uses a grasper and the ultrasonic shears to create windows in the gastrocolic ligament (Fig. 16.2). It is advisable to start outside the gastroepiploic arcade and divide the arcade at the end of the dissection. The first step to follow is the mobilization of the greater curvature followed by the second step of dissection of the antrum and inferior aspect of the duodenum. At this point the right gastroepiploic artery is divided between clips rather than applying electrocautery or using the ultrasonic shears alone.

Using a right-angled dissector, exactly as it is used for dissection of the esophagus, a retroduodenal passage is created starting at the inferior aspect of the duodenum. Dissection then proceeds to the superior aspect of the duodenum, and the right gastric artery is ligated between clips and divided. Again, the right-angled 10-mm dissector is introduced into the subxyphoid port to complete the dissection behind the duodenum, as this port is immediately in line with the dissection (Fig. 16.3). When the retroduodenal space has been created, an umbilical tape is passed around the duodenum, and the window is enlarged to introduce a 60-mm linear cutter through the same subxyphoid port in the same direction to perform the transection. Blue loads are typically used, but green loads can be used if the duodenum is thickened. After dissection the stomach can be pulled upward and the lesser curvature is skeletonized. The posterior attachments of the stomach to the pancreas are divided to allow full mobilization of the stomach. Proximal resection is also performed using an endoscopic linear stapler to the previous dissection limits of the lesser and the greater curvature.

Figure 16.2 Initiation of the gastrectomy. An asterisk marks the beginning of the dissection (division of the gastroepiploic arcade).

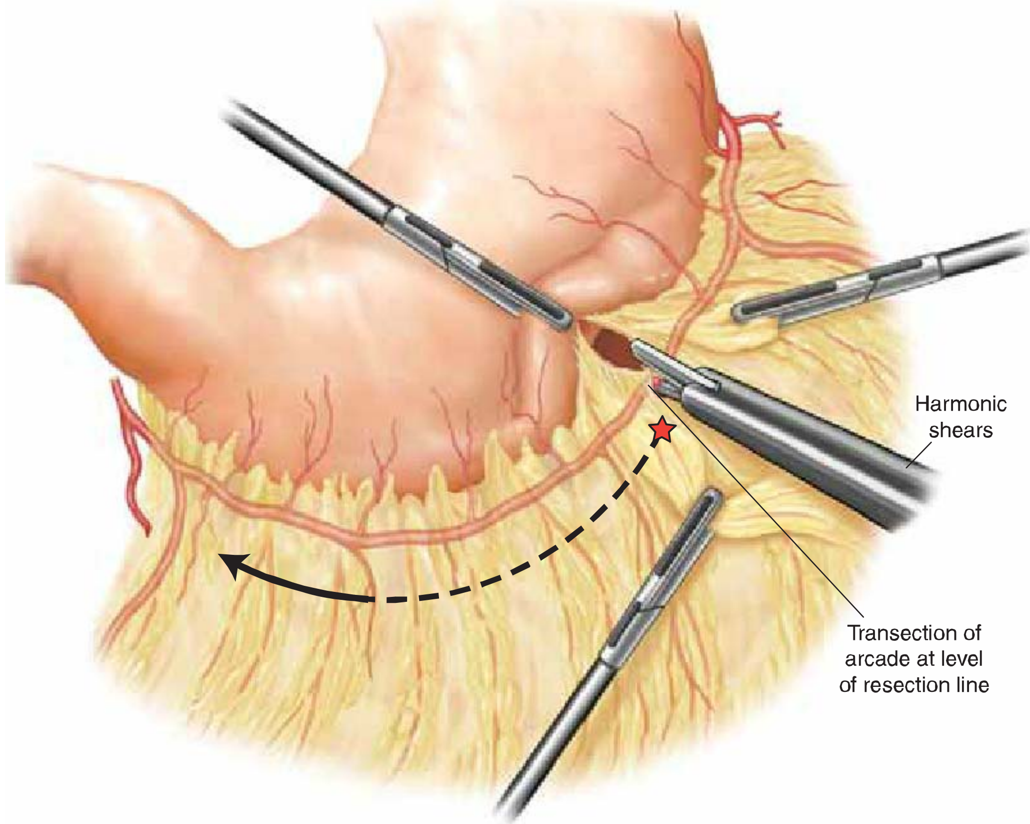
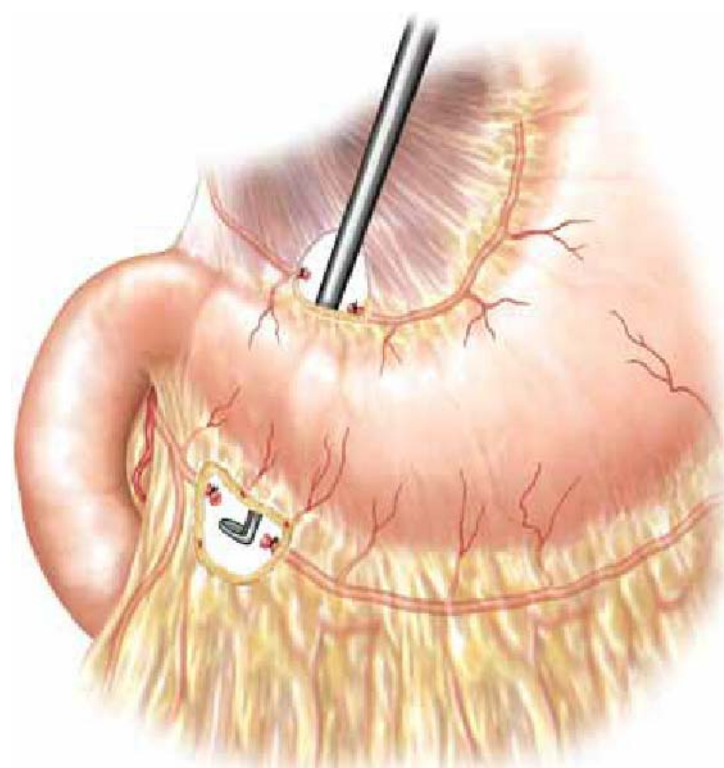


Figure 16.3 Creation of a retroduodenal passage using the subxyphoid port.



Alternative Lesser Curvature First Technique

The gastrohepatic ligament is divided to enter the lesser sac. The proximal extent of the dissection is the lesser curvature incisura where the left gastric vessels enter the stomach. The distal extent of the dissection is the pylorus, ligating and dividing the right gastric vessels.

The omentum is lifted up and its attachment to transverse colon and mesocolon is divided to enter the lesser sac inferiorly. Dissection is carried proximally to the fundus of the stomach ligating approximately the distal half of the short gastric vessels along the greater curvature including all lymph nodes. The right gastroepiploic vessels should be identified and ligated using a linear cutter stapler with vascular loads, and any posterior adhesions from stomach to pancreas should be divided.

Distal resection is done using a linear cutter stapler at the duodenal bulb just distal to the pylorus. If the tumor abuts the pylorus, the first portion of the duodenum should be taken to the point where it is safe and the margins should be checked with frozen sections. The duodenal stump is not routinely oversewn. The proximal resection is also performed using a linear cutter stapler to the previous dissection limits of the lesser and the greater curvature. Every attempt should be made for an R0 resection.

D1-Lymphadenectomy

The resection includes the removal of the specimen en bloc with the omentum and the regional perigastric lymph nodes stations 1–6, which are around the lesser and greater curvature, as well as the removal of the nodes along the left gastric artery (station 7). It does not include lymph node removal along the common hepatic artery, celiac axis, splenic artery, and proper hepatic artery.

Specimen Retrieval

The camera port site is enlarged to approximately 5 cm, a wound protector is placed, and the gastrectomy specimen is removed and inspected grossly. The size of the incision can be varied depending on the size of the tumor. Frozen sections can be done as needed of the proximal, distal, or radial margins. The proximal jejunum is eviscerated and is divided using the linear cutter stapler between 10 and 20 cm from the ligament of Treitz where the mesentery will reach the remnant stomach without tension in the antecolic position. Then a stapled side to side jejunojejunostomy is created between the biliary limb and the distal jejunum leaving approximately 50 cm of the alimentary limb in order to prevent bile reflux. If feeding jejunostomy tube is indicated, it is placed in the alimentary limb in an appropriate position to avoid tension in the yet to be made proximal gastrojejunostomy anastomosis.

Anastomosis

Gastrojejunostomy can be made by hand-sewn sutures or circular EEA staplers, both done intracorporeally. In order to achieve pneumoperitoneum again, a gel port can be attached to an appropriately sized wound protector. It is difficult to prevent all leakage of pneumoperitoneum, but enough space can be achieved for adequate anastomosis.

Hand-sewn Anastomosis

- End to side gastrojejunostomy in single layer using absorbable sutures such as 3-0 Polydioxanone (Ethicon, Somerville, NJ) after lining them up in the two corners.
- Anastomosis should be 6 cm in length.
- An alternative technique is to reconstruct the gastrojejunostomy with a 70-cm Roux-en-Y limb, in the same technique as a hand-sewn laparoscopic gastric bypass technique.

Stapled Anastomosis

- A circular stapler can be used for anastomosis. EEA with a 25-mm OrVil (Covidien, Norwalk, CT) , which is the largest size made, is used. The tube end of the OrVil is introduced transorally prior to enlarging the incision. This is done to minimize the pneumoperitoneum time after the manipulation of the incision when leak is common.

The OrVil should be well lubricated and the anesthesiologist is asked to pass it transorally until the tip can be seen inside the remnant stomach. A small gastrotomy is made to allow passage of the tube which is pulled and guided gently out of a working port while the anesthesiologist guides the well lubricated anvil into the mouth. The tube is pulled until the post of the anvil is out of the gastrotomy and the suture is seen. The suture is cut, which straightens the previously flat anvil. Then the anvil is left in this position until the time of the anastomosis. The EEA stapler is introduced through the gel port. The staple line of the proximal end of the alimentary limb is excised and the EEA stapler is placed inside. Then it is advanced approximately 5 cm, and the spike is opened at the anti-mesenteric border. The anvil and the stapler are mated carefully to avoid any twisting of the jejunum, and the stapler is fired creating the anastomosis. The EEA with the anvil is removed with gentle twisting motion, and the opening at the end of the hockey stick is closed using a linear cutter stapler.

- A linear stapler can be used for anastomosis. The anastomosis can be performed on the anterior or posterior aspect of the stomach (Fig. 16.4). After performing an enterotomy on the jejunal limb and the remnant stomach, it is advisable to use several firings of a

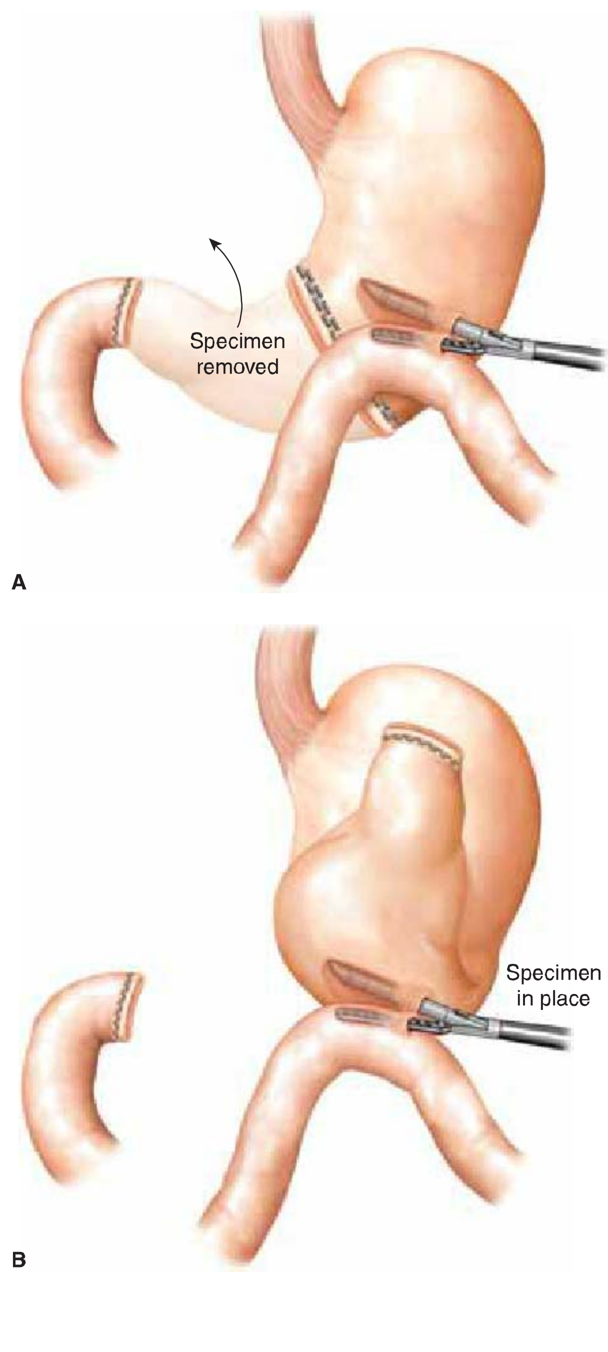


Figure 16.4 Intra-abdominal Billroth II reconstruction: (**A**) resection of the specimen first; (**B**) Billroth II reconstruction first, specimen in place.

30-mm linear cutter rather than a single firing of a 60-mm linear cutter which is bulky and more difficult to handle in this instance. Green staples are preferable if the stomach is thickened. After stapling, the enterotomies are closed using intracorporeal suturing techniques with a running 3-0 Prolene suture. This is preferable to the application of linear cutters that could narrow the anastomosis. It is also advisable to leave the nasogastric tube in the jejunal loop to calibrate the loop and avoid any bites in the posterior wall while suturing the gastrotomies and enterotomies.

OTHER TECHNICAL NOTES

One Jackson Pratt drain is placed in a dependant position to run adjacent to the gastrojejunostomy anastomosis and the duodenal stump. Omentectomy and regional lymphadenectomy are omitted if resection is done for benign disease. Adjacent organs are resected as needed en bloc if either gross inspection or frozen section reveals extension. At the end of the procedure there are two important final steps: First, a leak test should be performed with methylene blue and/or air insufflation of the stomach; second, the patient should be placed in Trendelenburg position to remove all fluid with suction, as some enteral fluid may remain otherwise and promote abscess formation in the postoperative period.

POSTOPERATIVE MANAGEMENT

Routine postoperative prophylaxis against deep vein thrombosis and atelectasis includes early ambulation and incentive spirometry.

Routine use of nasogastric tube is controversial, but we place it routinely with the tip across the anastomosis, and it is connected to low continuous wall suction until the output is minimal, which usually takes 1 to 2 days.

Contrast studies to assess for leak are not routinely used in subtotal gastrectomies unless it is clinically indicated. Jejunostomy tube, if placed, needs routine flushing in order to prevent clogging, and tube feeding is started on day 2 and used in the immediate postoperative period as nutritional supplementation until patients are able to take in adequate nutritional requirements orally. It can be removed after 4 weeks if it is not needed but should be left in longer if the patient is receiving adjuvant therapy, so it can be used if the patient experiences nausea from the medications. If the feeding tube gets clogged or is removed accidentally, it can be replaced with a wire and fluoroscopic guidance by interventional radiology.

The Jackson Pratt drain is removed when output is low after the patient is started on a diet.

The patient should be followed by the oncologist for any adjuvant therapy, surveillance, and vitamin B12 supplementation.

COMPLICATIONS

Patients generally recover quickly from the procedure with low complication rates. Anastomotic leak rate is very low at 1 to 2% but can be higher in irradiated or malnourished patients. Patients with a jejunostomy tube can rely on tube feeding for all of their nutritional requirements while the leak heals. Duodenal stump leak or other leak from jejunojejunostomy or feeding jejunostomy site is extremely uncommon but should be suspected if the patient has peritonitis or has enteric output from the drains. Undrained leaks can be treated by percutaneous drain placement in stable patients or operative drainage in sick patients.

Wound infections of the mini-midline wound occur more frequently in patients who are immunocompromised and in grossly contaminated cases. Wound protector usage has not been associated with complete elimination of infections. The overall postoperative morbidity rate is around 10% and mortality rate approaches 0 to 1%.

RESULTS

In experienced hands, the laparoscopic approach shows similar results to the open approach with similar number of lymph nodes harvested and overall survival.

CONCLUSIONS

Laparoscopic subtotal gastrectomy with regional lymphadenectomy should be the procedure of choice in patients with resectable gastric cancer. Novel minimally invasive approaches such as natural orifice translumenal endoscopic surgery, single-incision laparoscopy, sentinel node biopsy, and robotic operations have been reported but the benefits are yet to be determined.

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17 Subtotal Gastrectomy and D2 Resection

Carmine Volpe and Bestoun H. Ahmed

INDICATIONS/CONTRAINDICATIONS

The worldwide geographical distribution of gastric carcinoma is variable. It is more common in Japan and other far eastern countries. It is the second worldwide leading cause of cancer death. In the west the incidence of distal gastric cancers has been decreasing for decades; however, cancer of esophagogastric junction and cardia is increasing. Histologically, gastric cancer can be divided into two types, intestinal type and diffuse type, according to Lauren classification. The intestinal type is more common in high-incidence countries like Japan, more likely found in the distal stomach and associated with *Helicobacter pylori*. The diffuse type is more common in young patients, in hereditary gastric cancer, in proximal locations, and is associated with a worse prognosis. Distal subtotal gastrectomy (50% to 75%) is the recommended surgical treatment for distal cancer of the stomach with fewer complications, shorter hospital stays, better quality of life, and no difference in overall survival compared to total gastrectomy.

The east and west remain sharply divided regarding the extent of lymph node dissection (D1, D2) for gastric adenocarcinoma. The general rules of the Japanese Research Society for Gastric Cancer divide the anatomic perigastric lymph nodes into 16 groups (Table 17.1). The anatomic location of the perigastric and extragastric lymph nodes are shown in Figure 17.1. The nodal groups are further subdivided into three regions (N1, N2, and N3), and their designations vary depending on whether the gastric cancer is located in the proximal, middle, or distal third of the stomach. The N1 and N2 lymph node groups of the distal third of the stomach are illustrated in Table 17.2. Removing only the N1 lymph nodes constitutes a D1 gastric resection while the more extended D2 resection requires the removal of both N1 and N2 lymph node groups. The current pathologic N stage is defined by the number of metastatic lymph nodes and not their anatomic location relative to the primary tumor. The American Joint Committee on Cancer recommends that retrieval of at least 15 lymph nodes is necessary to accurately stage a gastric cancer.

TABLE 17.1 Perigastric and Regional Lymph Node Groupings

Location	Group no.
Right cardiac	1
Left cardiac	2
Lesser curvature	3
Greater curvature	4
Suprapyloric	5
Infrapyloric	6
Left gastric artery	7
Common hepatic artery	8
Celiac artery	9
Splenic hilum	10
Splenic artery	11
Hepatoduodenal ligament	12
Retropancreatic	13
Base of SMA	14
Middle colic artery	15
Para-aortic	16

From Japanese Research Society for Gastric Cancer: The general rules for gastric cancer study in surgery and pathology. *Jpn J Surg* 1981;111:127.

TABLE 17.2 Lymph Node Groups Resected for Distal Gastric Cancers According to Extent of Resection

D1 Gastrectomy	D2 Gastrectomy
No. 3—lesser curve	No. 1—right cardiac
No. 4—greater curve	No. 3—lesser curve
No. 5—suprapylorus	No. 4—greater curve
No. 6—infrapylorus	No. 5—suprapylorus
	No. 6—infrapylorus
	No. 7—left gastric
	No. 8—hepatic artery
	No. 9—celiac artery

From Japanese Research Society for Gastric Cancer: The general rules for gastric cancer study in surgery and pathology. *Jpn J Surg* 1981;111:127.

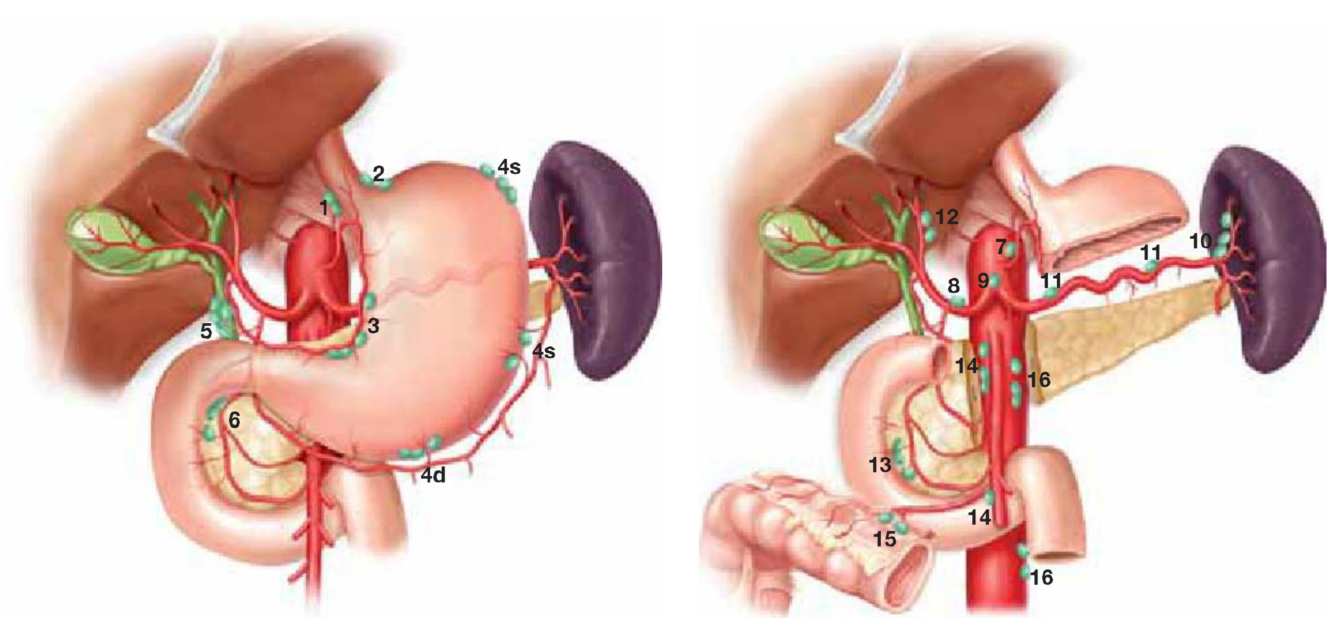

Figure 17.1 Lymph node locations according to Japanese Research Society for Gastric Cancer. **A:** D1 lymph node stations. **B:** D2 lymph node stations.

Indications for Distal Subtotal D2 Gastrectomy

- Patients should be medically fit.
- Adenocarcinoma of the distal third of the stomach
- Gastric tumor that does not cross the line extending from the bare area on the greater curvature (between portion of the stomach supplied by the gastroepiploic artery and the area supplied by the short gastric) to a point on the lesser curvature 5 cm below the cardioesophageal junction.
- Intestinal-type gastric cancer according to the Lauren classification
- T3/T4 tumors after neoadjuvant therapy

Contraindications

- N3 lymph node involvement
- Diffuse type histology
- Cancers within 5 cm of the gastroesophageal junction
- M1 disease

PREOPERATIVE PLANNING

The goals of preoperative planning and assessment are to obtain a tissue biopsy for diagnosis, stage the extent of disease, and to address the patient's nutritional status and significant medical comorbidities prior to surgery. Tumor-related complications such as, gastric outlet obstruction, and bleeding may require hospital admission for resuscitation and replacement therapy prior to surgery.

Preoperative assessment includes the following:

- Fiberoptic esophagogastroduodenoscopy for diagnostic biopsy and to determine the location of the primary tumor in relation to the EG junction and pylorus
- To optimize tumor staging endoscopic ultrasonography has been utilized to assess tumor invasion into the gastric wall (T stage), invasion into adjacent organs such as, spleen, pancreas, and left lobe of the liver, and fine needle aspiration of perigastric lymphadenopathy.
- CT scan of the abdomen and pelvis to determine tumor resectability and tumor involvement of peritoneum, lymph nodes, and liver (M1).
- Diagnostic laparoscopy is recommended for patients with advanced tumors T3/T4 who are not obstructed or bleeding, patients considered at high risk for M1 disease and patients deemed candidates for neoadjuvant therapy.
- Nutritional support and nasogastric decompression for patients with gastric outlet obstruction
- Preoperative blood transfusions for patients with hemocrit <25%, hemoglobin <8 g
- Start a beta blocker 1 week prior to surgery for patients over 65-years old or those who have a significant cardiac history.
- Preoperative prophylactic heparin should be given after placement of epidural catheter.

SURGERY

The aim of surgical therapy is to resect the gastric tumor with negative margins (4 cm proximal, 2 cm distal) and to retrieve at least 15 lymph nodes for pathologic examination. The surgical specimen should include 50% to 75% of the stomach en bloc with the greater and lesser omentum, lesser sac bursa, lymph nodes from the lesser, and greater curvature of the stomach (N1), from above and below the pylorus (N1), right cardia, celiac axis, and its branches (N2). Since splenectomy or pancreatectomy does not offer a survival advantage the recommended procedure is a D2 lymph node dissection with adjacent organ preserving distal subtotal gastrectomy.

Positioning

The patient should be placed in the supine position with the arms abducted. Reverse Trendelenburg positioning will facilitate exposure. After general anesthesia is obtained via endotracheal intubation a nasogastric tube should be placed in the stomach for decompression. Place sequential compression devices on the lower extremities.

Technique

A variety of standard abdominal incisions are available that provide adequate exposure for gastric resections; we prefer a bilateral subcostal incision (Chevron) because this affords wide exposure of the upper abdomen and excellent retraction of the costal margin.

Diagnostic laparoscopy is indicated prior to incision if advanced disease is suspected.

Upon entering the abdomen, careful exploration should be performed looking for gross metastatic disease. The location and mobility of the primary tumor and the need for bloc resection of adjacent organs and structures are evaluated. A small aliquot of the abdominal fluid is aspirated and sent for cytologic evaluation. A mechanical retractor is placed in the incision to elevate the costal margin and retract the liver.

- The procedure begins with omentobursectomy, the greater omentum is detached from the transverse colon along with the anterior leaf of the transverse mesocolon (Fig. 17.2).
- With cephalad retraction of the greater omentum an avascular plane is entered just above the transverse colon. Bleeding at this point indicates that dissection is too deep into the transverse mesocolon proper (Fig. 17.3).
- Dissection proceeds over the anterior aspect of the pancreas. The pancreatic capsule is opened and the anterior layer of the capsule which is part of the lesser sac is mobilized toward the specimen up to the common hepatic artery.

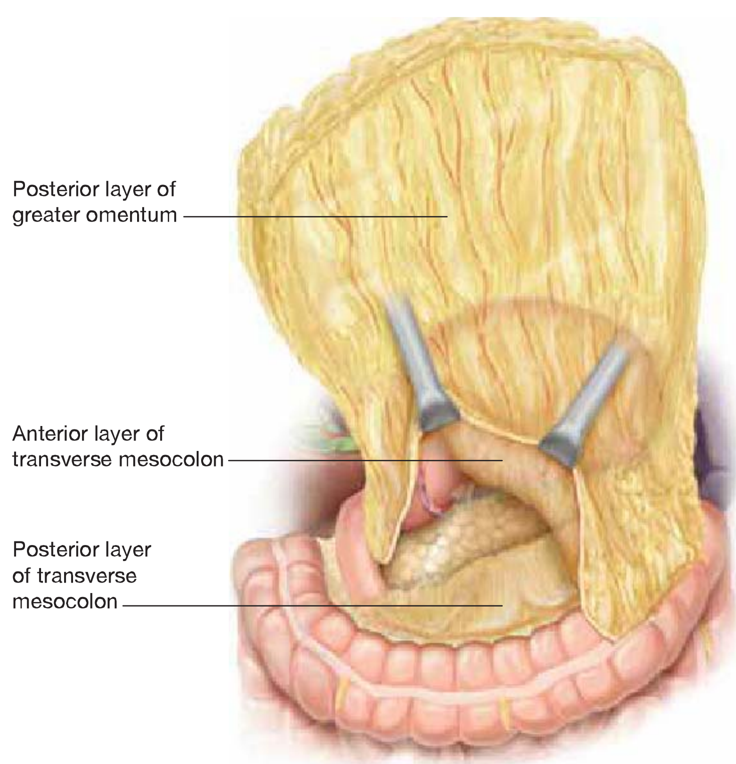

Figure 17.2 Lesser bursectomy, dissecting en bloc the anterior sheet of the transverse mesocolon and anterior capsule of the pancreas along with its associated lymphatics.

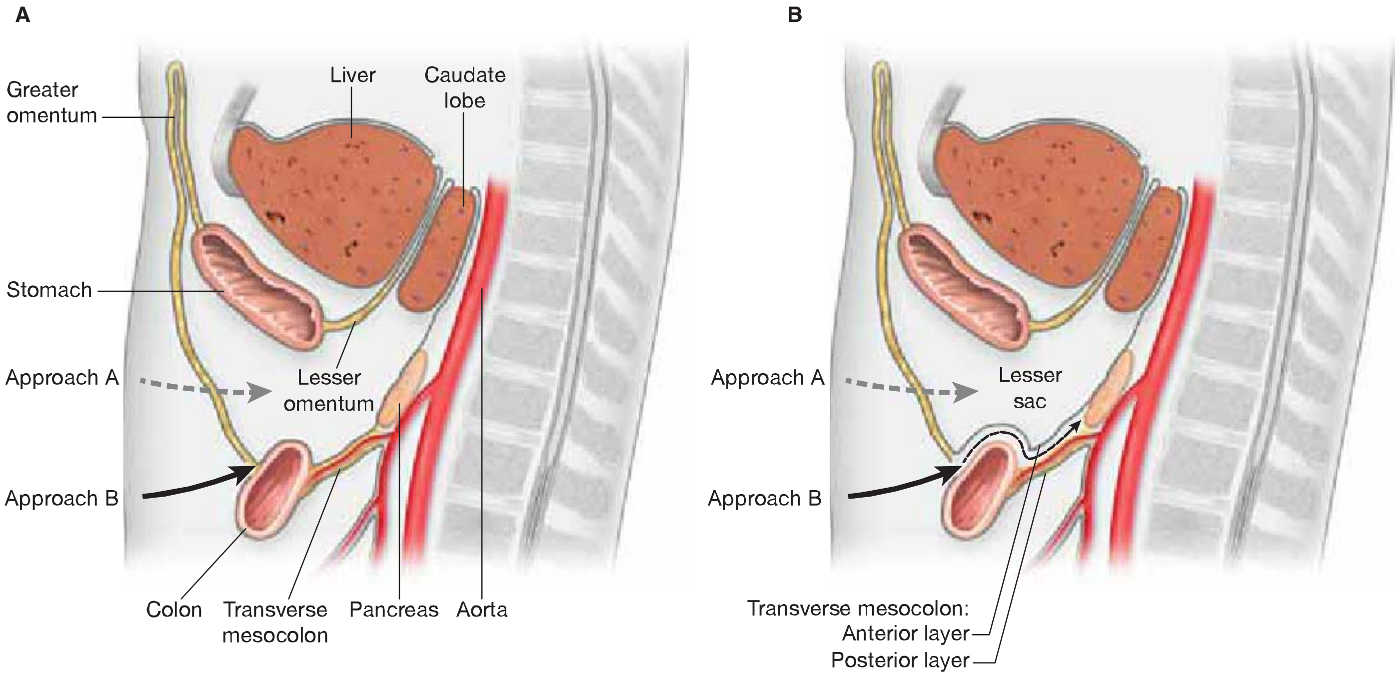

Figure 17.3 Proper plane of dissection for complete lesser bursectomy. **A:** Approach "A" demonstrates plane of entry that is too shallow, which would leave behind the posterior aspect of the lesser sac. Approach "B" is the proper plane of entry, which is deep to the anterior leaf of the transverse mesocolon. **B:** Once the proper plane is entered as shown by approach "B" the posterior aspect of the lesser sac is removed along with specimen allowing complete bursectomy.

- Continue dissecting in this plane toward the superior margin of the pancreas; the right gastroepiploic vessels are identified, clamped divided, and ligated; and the infrapyloric lymph nodes 6 are dissected toward the specimen (Fig. 17.4).
- Perform a Kocher maneuver elevating the duodenum out of the retroperitoneum and dissecting the first portion of the duodenum off the head of the pancreas.
- *Lesser omentectomy:* Incise the triangular ligament of the left lobe of the liver with electrocautery avoiding the left phrenic vein. Elevate the lateral segment of the left lobe to expose the attachment of the lesser omentum to the ductus venosum on the

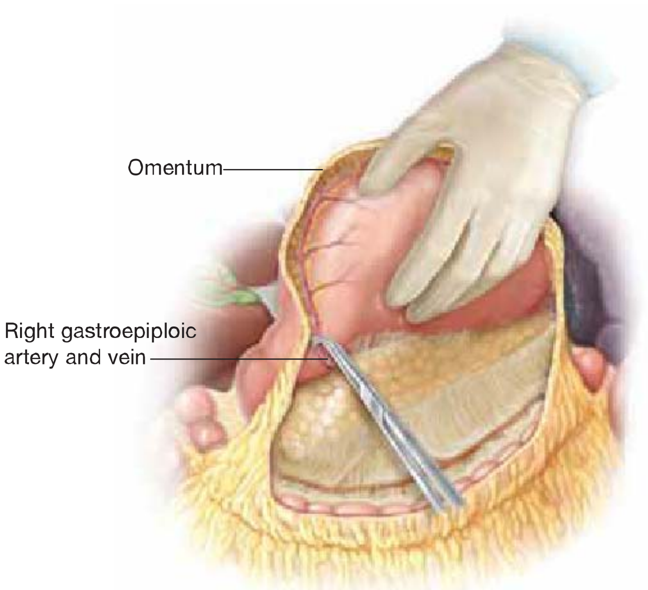

Figure 17.4 Ligation of the right gastroepiploic vessels at the superior border of the pancreas near its origin off the gastroduodenal artery.

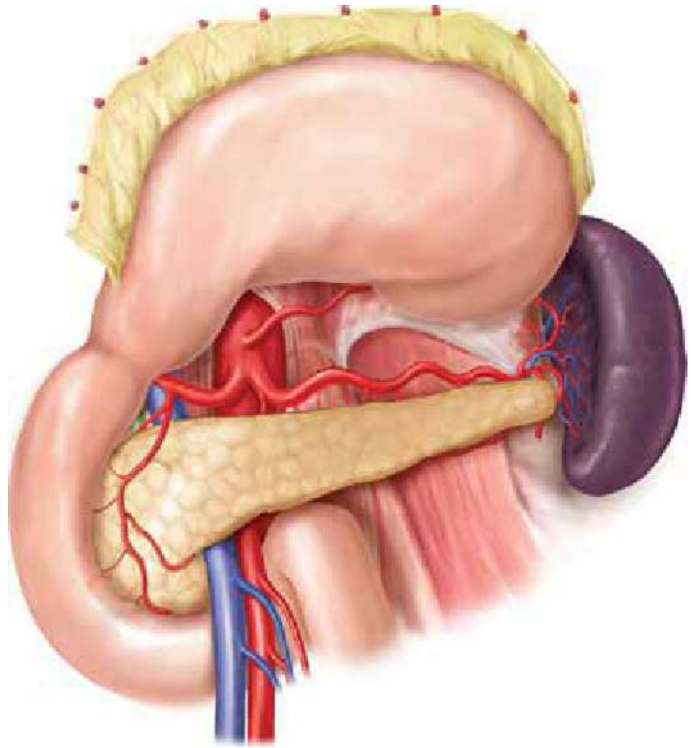

Figure 17.5 Stomach reflected superiorly displaying the celiac axis.

undersurface of the liver running from the left portal vein to the left hepatic vein. Detach the lesser omentum from the ductus venosum with electrocautery along its entire length over the caudate lobe from the right paracardial region down to the anterior leaf of the hepatoduodenal ligament where the right gastric artery is identified. Clamp divide and ligate the right gastric artery at its origin from the common hepatic artery. This ensures complete removal of the lesser sac bursa.

- Complete circumferential dissection of the first portion of the duodenum by ligating supraduodenal arterial branches off the gastroduodenal artery and transect the duodenum with a linear stapler (blue load) approximately 2 cm distal to the pylorus. Request frozen section to confirm a negative surgical distal margin. These maneuvers will ensure that lymph node groups 3 and 5 are included with the specimen.
- *Celiac axis dissection:* The divided stomach is reflected superiorly and to the left (Fig. 17.5). Dissection proceeds along the superior border of the pancreas and along the common hepatic artery toward the celiac axis, including lymph nodes 8 and 9. The left gastric artery is ligated and divided at the celiac axis removing lymph nodes 7. For 75% distal subtotal gastrectomy: dissect the splenic artery near its origin from the celiac axis toward the splenic hilum sweeping lymph nodes 11 toward the specimen.
- With the specimen retracted superiorly, the area of proximal gastric resection on the greater curvature is determined with the goal of achieving a 4-cm proximal margin. The greater omentum is liberated from the greater curvature at this point and one or two short gastric vessels are ligated and divided as necessary to free this area. In order to maintain the viability of the gastric remnant some of the short gastric vessels must be preserved when the left gastric artery is taken at its origin.
- Proximal gastric resection is complete with a TA-90 or GIA linear green stapler in a manner to ensure complete removal of a rim of lesser curvature up to the cardia. Frozen section analysis is obtained to confirm a histologic negative proximal margin.

Reconstruction

Our preference is to perform a retrocolic Billroth II reconstruction with a two-layer hand-sewn Hofmeister-type anastomosis. We prefer the retrocolic location due to the shorter distance to travel and posterior location behind the stomach.

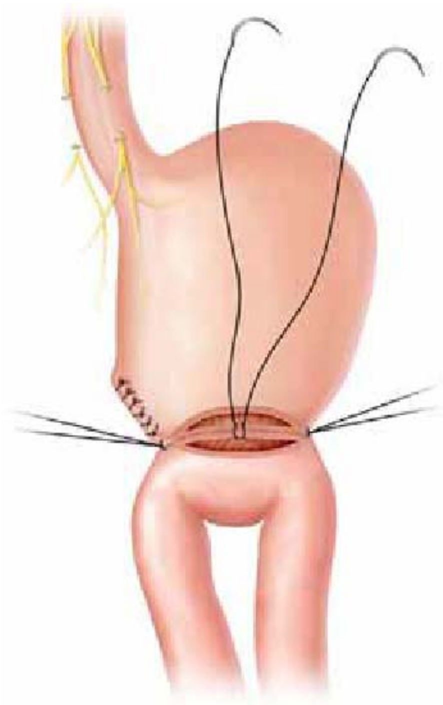

Figure 17.6 Closure of the posterior mucosal layer consists of two continuous sutures of absorbable material starting in the middle and running to each corner.

- The transverse colon is retracted upward and the transverse mesocolon inspected for the identification of an avascular area to the left of the middle colic vessels. A proximal loop of jejunum is delivered through this avascular window in the transverse mesocolon to lie opposed to the gastric remnant. The distance from the ligament of Treitz to the gastric remnant should be no more than 20 cm.
- Align the loop of jejunum up to the greater curvature of the gastric remnant.
- *Posterior serosal layer:* Place interrupted 3-0 silk seromuscular Lembert sutures about 5 mm apart between the posterior gastric wall and the antimesenteric border of the jejunum. Tag the corner sutures with a hemostat and cut the remaining silk tails. Sutures in the gastric wall should be placed about 1 cm from the gastric staple line.
- Excise approximately 3 cm of the gastric staple line from the greater curvature. Make a corresponding incision in the jejunum with electrocautery along the antimesenteric border about two-thirds the size of the gastric opening.
- *Posterior Mucosal layer:* Place two 3-0 absorbable sutures next to each other in the midpoint of the posterior layer inserted full thickness through the jejunal and gastric walls. Tie the sutures and then tie the tails together (Fig. 17.6). Run each suture from the midpoint in a continuous fashion to their corresponding corner. Sutures should be placed in a "V" configuration with a larger serosal bite compared to mucosa. The final corner suture should pass from inside to outside of the anterior gastric wall. This suture will be tied to the interrupted corner stitch from the anterior mucosal layer.
- *Anterior Mucosal Layer:* This is a full thickness layer closure with interrupted 3-0 absorbable sutures with knots tied outside the lumen. Place the corner sutures first next to the suture from the posterior mucosal layer. Tie the interrupted sutures to itself and then to the strand from the posterior mucosal layer. Sutures placed in this layer resemble an inverted "^" configuration. Prior to completing the anastomosis ask the anesthesiologist to place the nasogastric tube in the gastric remnant just above the anastomosis.
- *Anterior Serosal layer:* Place a row of interrupted 3-0 silk seromuscular Lembert sutures between the anterior gastric wall and the jejunum inverting or dunking the anterior mucosal layer.

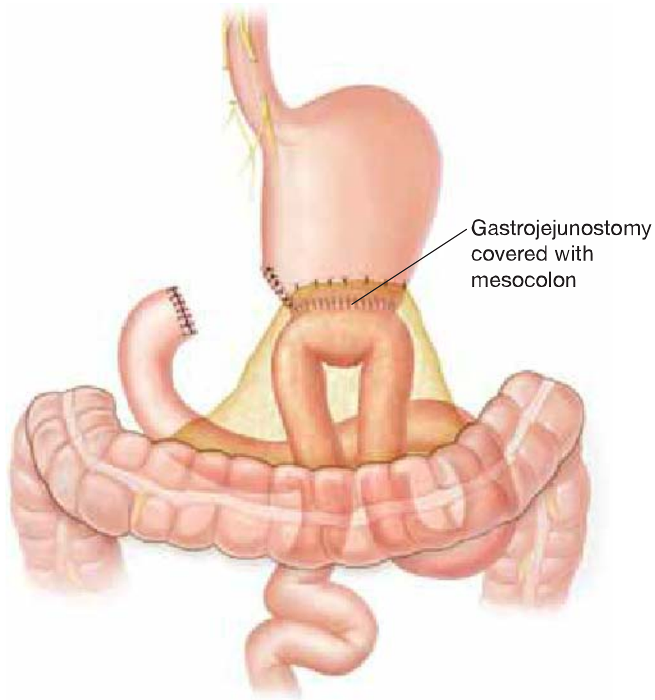

Figure 17.7 Completed Billroth II retrocolic gastrojejunostomy.

- To prevent tension on the anastomosis and internal herniation of small bowel through the opening in the transverse mesocolon, suture the jejunal loop to the peritoneum of the transverse mesocolon with interrupted 3-0 absorbable sutures being careful to avoid the mesenteric vessels. Figure 17.7 illustrates the completed anastomosis.
- We do not drain the anastomosis or the lesser sac routinely unless there is concern of a possible pancreatic injury or leak. If necessary we use a closed suction 10 French Jackson–Pratt flat drain exteriorized through a separate incision.
- The incision is closed in two layers, the posterior sheath with a nonabsorbable suture and anterior sheath with absorbable suture.
- The skin is approximated with skin staples.
- Alternatively, gastrojejunostomy Billroth II reconstruction can be created with GIA and TA stapling devices.
- Oppose the posterior gastric wall to the antimesenteric border of the jejunum with traction sutures. The anastomotic site on the posterior gastric wall is 2 to 3 cm proximal to the gastric staple line closure.
- Create a small gastrotomy on the greater curvature side and opposing enterotomy in the jejunum on the antimesenteric surface with electrocautery.
- Insert the GIA limbs into the gastric remnant and jejunum (Fig. 17.8).
- Engage the limbs of the GIA, and before deploying the device inspect to ensure that the serosal surfaces are directly apposed with nothing between them.
- Withdraw the limbs of the GIA and inspect the anastomosis for hemostasis.
- Close the resultant gastroenterotomy with a TA stapler placing the GIA staple lines at the corners to splay open the anastomosis (Fig. 17.9).

Laparoscopic Approach

- The patient is placed on the table in the supine split leg position. A slight reverse Trendelenburg (15 to 30°) position is very helpful. The surgeon alternates position in between legs and the patient's right side, and the camera assistant is on the left side.
- Pneumoperitoneum is established via Veress needle in the left upper quadrant or Hasson technique to reach an intra-abdominal pressure of 15 mm Hg.

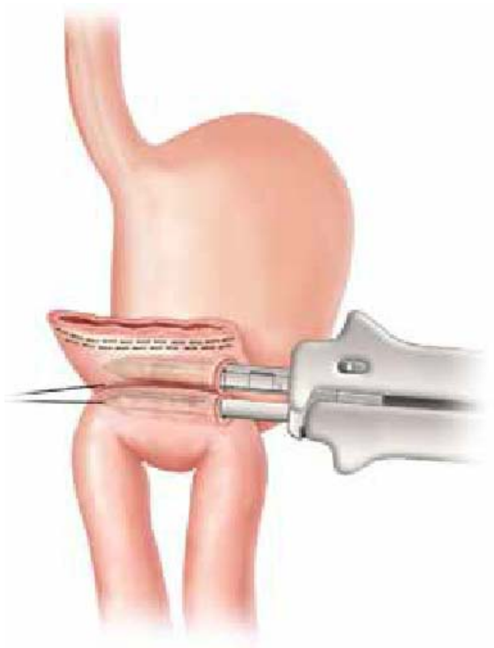

Figure 17.8 Stapled gastrojejunostomy. (Insertion of the GIA limbs into the stomach and jejunum.)

- Access to the abdominal cavity is obtained by the use of four 5-mm laparoscopic trocars and two 12-mm trocar. These ports are placed in a V-shaped arrangement around the umbilicus (Fig. 17.10).
- The gastrocolic ligament is divided using ultrasonic shears (Ultracision-Harmonic Scalpel, Ethicon Endo-Surgery Inc., Cincinnati, OH) along the border of the transverse colon, including the greater omentum in the resected specimen and allowing access to the lesser sac.
- Dissection continues toward the pylorus to include the infrapyloric nodes (group 6) and division of the right gastroepiploic artery at its origin off the gastroduodenal artery.
- Create a window in the hepatoduodenal ligament with electrocautery.
- Identify the right gastric artery; divide and ligate it near its origin off the common hepatic artery.

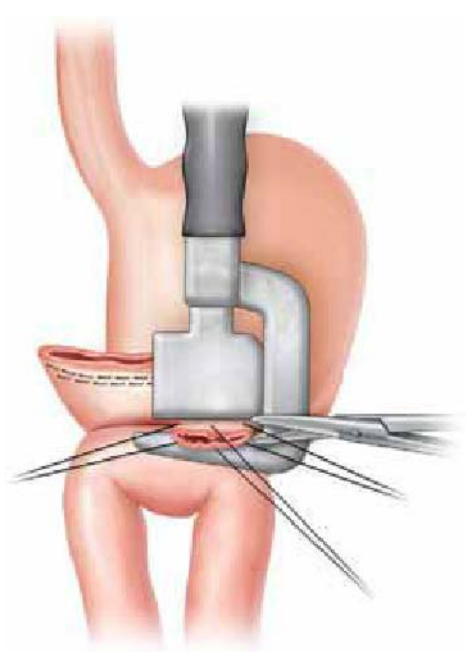

Figure 17.9 The anastomosis is completed with the application of a TA stapler.

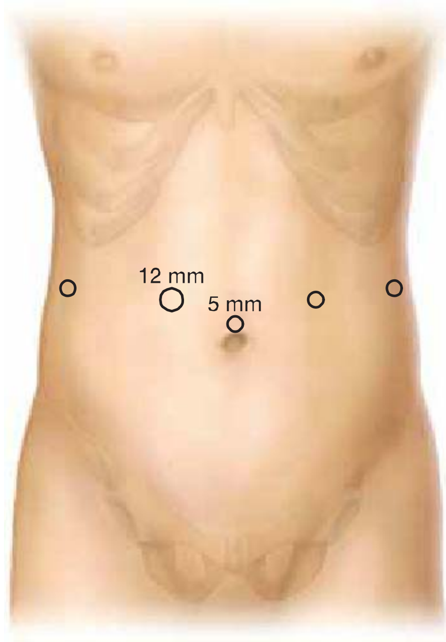

Figure 17.10 Laparoscopic port placement.

- Dissect the first portion of the duodenum off the head of the pancreas and divide it 2 cm distal to the pylorus with 35-mm linear endostapler.
- Retract the lateral segment of the left lobe of the liver and incise the lesser omentum from the undersurface of the liver up to the right paracardial region exposing the common hepatic artery and celiac axis.
- Continue dissection along the common hepatic artery toward the celiac axis. Identify the left gastric artery at its origin from the celiac artery and divide it. If necessary dissect the splenic artery from its origin toward the splenic hilum sweeping the lymph nodes toward the specimen (lymph node groups, 7, 8, 9, 11).
- Retract the stomach superiorly and determine the site of proximal resection along the greater curvature. Dissect this area free of the greater omentum and take 1 or 2 short gastric vessels if necessary.
- Resect the stomach with a 60-mm linear Endo-GIA stapler (green load) with the goal of achieving a 4-cm proximal margin and removing most of the lesser curvature (group 1 lymph nodes).
- Gastrojejunal anastomosis is performed using Billroth II antecolic technique. Bring a loop of jejunum anterior to the transverse colon up to the gastric remnant. The afferent limb should be no more than 20 cm in length beyond the ligament of Treitz.
- Place a posterior layer of seromuscular running absorbable suture to hold the stomach and jejunum in apposition.
- With electrocautery make a small gastrotomy along the greater curvature of the stomach 2 cm proximal to the gastric staple line closure.
- Make a corresponding enterotomy on the antimesenteric side of the jejunum.
- Insert the limbs of the 35-mm linear GIA stapler into the stomach and jejunum
- Engage the limbs of the device and fire the instrument.
- Close the GIA defect with an interrupted layer of absorbable sutures followed by a running outer layer of seromuscular Lembert silk suture.
- The specimen is retrieved through an enlarged 12-mm port-site incision via an Endocatch bag. A small wound protector should be used. The specimen retrieval could be delayed to the end of operation provided it is placed in the endobag.
- Close fascial wounds more than 5 mm in size with an absorbable suture followed by subcuticular skin closure.
- Laparoscopic D2 lymphadenectomy is feasible. Use of the robot, Da Vinci® Surgical System (Intuitive Surgical Inc., Sunnyvale, CA) is very helpful in this context secondary to achieving the most flexible and delicate intracorporeal laparoscopic movements.